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The co-evolution of direct, indirect and generalized reciprocity

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Abstract

People often engage in costly cooperation, especially in repeated interactions. When deciding whether to cooperate, individuals typically take into account how others have acted in the past. For instance, when one person is deciding whether to cooperate with another, they may consider how they were treated by the other party (direct reciprocity), how the other party treated others (indirect reciprocity), or how they themselves were treated by others in general (generalized reciprocity). Given these different approaches, it is unclear which strategy, or more specifically which mode of reciprocity, individuals will prefer. This study introduces a model where individuals decide how much weight to give each type of information when choosing to cooperate. Through equilibrium analysis, we find that all three modes of reciprocity can be sustained when individuals have sufficiently frequent interactions. However, the existence of such equilibria does not guarantee that individuals will learn to use them. Simulations show that when individuals mainly imitate others, generalized reciprocity often hinders cooperation, leading to defection even under conditions favorable to cooperation. In contrast, when individuals explore new strategies during learning, stable cooperation emerges through direct reciprocity. This study highlights the importance of studying all forms of reciprocity in unison.

INTRODUCTION

Reciprocity is one of the primary reasons why people cooperate with others (1). The promise of a reciprocal act often drives us to help others, even when it is costly to do so. However, the concept of reciprocity extends beyond direct reciprocity, the simple give and take between two individuals. People can also reciprocate by helping those who help others, even without benefiting directly (2–6), or by extending help to others after having received help themselves (7–9). These alternative forms of repayment are referred to as downstream and upstream reciprocity, or simply as indirect (10–13) and generalized reciprocity (10, 14–16).

Although the three kinds of reciprocity—direct, indirect, and generalized—are conceptually different, they are intertwined in practice. For example, if Alice has a positive experience with Bob but also notices that he defects with others, should she employ direct reciprocity (to cooperate) or indirect reciprocity (to defect) with him the next time they meet? Similarly, if Bob observes Alice defecting with others and is therefore unwilling to cooperate with her (as per indirect reciprocity), should he change his mind if Charlie just cooperated with him and he is in a positive mood (as per generalized reciprocity)? Most previous studies are unable to formally test these types of questions, either because they only study one kind of reciprocity (6, 12, 14, 17–22), or each kind is studied in isolation (23–26).

By now, experimental (27) and theoretical papers (28–30) have studied how individuals weigh between direct and indirect reciprocity. The broad conclusion from these studies is that people often integrate their own direct experience and the indirect experience of others to decide whether to cooperate with someone (27). In addition, there are situations where individuals predominantly prefer direct over indirect reciprocity. For example, when outside information is untrustworthy, individuals tend to ignore it and rely on their direct experience (28–30). These results on direct and indirect reciprocity lead up to the natural follow-up question: how do we integrate our general experience with others when we decide to cooperate with someone? In particular, are there situations in which individuals may prefer to cooperate via generalized reciprocity over the two other modes? Most theoretical models of generalized reciprocity either treat it as an isolated form of reciprocity (14, 15, 18, 31), or are, by construction, unable to study the competition between the different modes of reciprocity (32, 33). To address the questions posed above, however, a unified model is crucial.

In this paper, we introduce a model in which players can adopt strategies that cooperate through direct, indirect, or generalized reciprocity, or a combination of these mechanisms. More precisely, while interacting repeatedly within a population, individuals cooperate based on whether they per-

ceive their current co-player to be in good or bad standing. These standings are updated based on strategies that assign different weights to direct, indirect, and generalized experiences. Our model is a direct extension of the framework introduced by Schmid et al. (28), where they consider direct and indirect reciprocity in a unified model. Schmid et al. were the first to propose a framework that is explicitly analyzable and allows for analytical conclusions when comparing direct and indirect reciprocity. Similarly, our model remains mathematically tractable. We also extend the theory of zero-determinant strategies from direct reciprocity to a framework that includes indirect and generalized reciprocity, enabling us to derive analytical conclusions when comparing the three mechanisms.

RESULTS

A model of direct, indirect and generalized reciprocity

We consider a population of n individuals who repeatedly interact in pairs. More specifically, each interaction involves two randomly selected players from the population, who simultaneously and independently decide whether to cooperate or defect. If both cooperate, they incur a cost $c > 0$ to provide a benefit $b > c$ to their co-player. If they defect, they pay no cost and provide no benefit. All individuals not participating in the current interaction observe its outcome. However, these third-party observations are subject to errors: each observer independently mistakes cooperation for defection (or *vice versa*) with probability ε. We assume that players do not make observation errors in interactions they are directly involved in. Once the interaction concludes, another one occurs with probability d and with probability $1 - d$, the game ends. In the next interaction, two individuals are again randomly paired to interact in the same manner as before. The game's setup is illustrated in Fig. 1A. At the end of the game, players' payoffs are calculated by averaging over all the interactions in which they participated.

The decision to cooperate or defect in an interaction depends on how players currently perceive their co-player. Players can view their co-players in one of two binary standings: "good" or "bad". If a player perceives their co-player as good, they cooperate; otherwise, they defect (Fig. 1B). These interpersonal standings are referred to as reputations. Reputations are updated throughout the game based on the strategies of individuals (Fig. 1C). A player's strategy, denoted as $\sigma = (y, p, q, \lambda, \gamma)$, is a tuple with five components. The parameter y represents the initial probability that a co-player is deemed good in the absence of prior information. This applies when the player is interacting with someone for the first time and has yet to form an opinion. The strategy components p and q determine the likelihood of assigning a good reputation to a co-player. Specifically, p represents

the probability of assigning a good reputation after cooperation, while q represents the probability after defection.

Besides direct interactions, players may also update reputations based on indirect information. If a co-player interacts with a third party, the focal player updates that co-player's reputation with probability λ (indirect reciprocity). In this case, the co-player is assigned a good reputation with probability p (or q), if the co-player cooperated (or defected). In addition, after a direct interaction with a third party, the focal player herself updates the reputation of every uninvolved co-player, independently, with probability γ (generalized reciprocity). Once again, reputations are updated according to p or q based on whether the focal player's interaction partner cooperated or defected. For a more detailed description of the strategies, see Section 1 of the **Supplementary Information**.

Let us now consider the extreme cases of the model. When $\lambda = \gamma = 0$, players rely exclusively on direct reciprocity, meaning reputations are updated solely based on their direct interactions with co-players. In this case, reputations change only according to the outcomes of a player's personal experiences. If $\lambda = 0$ and $\gamma = 1$, reputations are still updated based on the player's own interactions, but the update does not involve the co-player directly. Instead, players use their most recent interaction to update the reputation of another individual. This scenario represents pure generalized reciprocity. With $\gamma = 0$ and $\lambda = 1$, players update the reputation of a third party whenever that individual participates in an interaction, irrespective of the observer's direct relationship with them. This process, which relies on interactions involving others, is referred to as pure indirect reciprocity. Players combine the three modes of reciprocity, direct, indirect and generalized when $0 < \lambda < 1$ and $0 < \gamma < 1$.

Note that, for $\gamma = 0$, the model reduces to the model studied by Schmid et al. (28), where no generalized reciprocity occurs. Our framework is a natural extension of their work, incorporating generalized reciprocity to provide a more unified framework of reciprocity. Similar to the previous model, it can encompass several well-known strategies from both the direct and indirect reciprocity literature, such as Tit-for-Tat (TFT), Generous Tit-for-Tat (GTFT), and the simple scoring strategy from indirect reciprocity (34), which assigns good reputations to cooperators and bad reputations to defectors, while always cooperating in anonymous interactions. These strategies are represented as $(1, 1, 0, 0, 0)$, $(1, 1, q, 0, 0)$, and $(1, 1, 0, 1, 0)$, respectively, using the current notation. Additionally, our framework can also represent strategies from the generalized reciprocity literature, such as Alternating Tit-for-Tat (A-TFT) (14) or Upstream-TFT (23), denoted as $(1, 1, 0, 0, 1)$. A-TFT is a strategy that repeats the last action of their last co-player in the next interaction.

Our model has two salient properties: (i) conditional cooperation by an individual depends solely on the current standing of their co-player, and (ii) any reputation update is based on a single

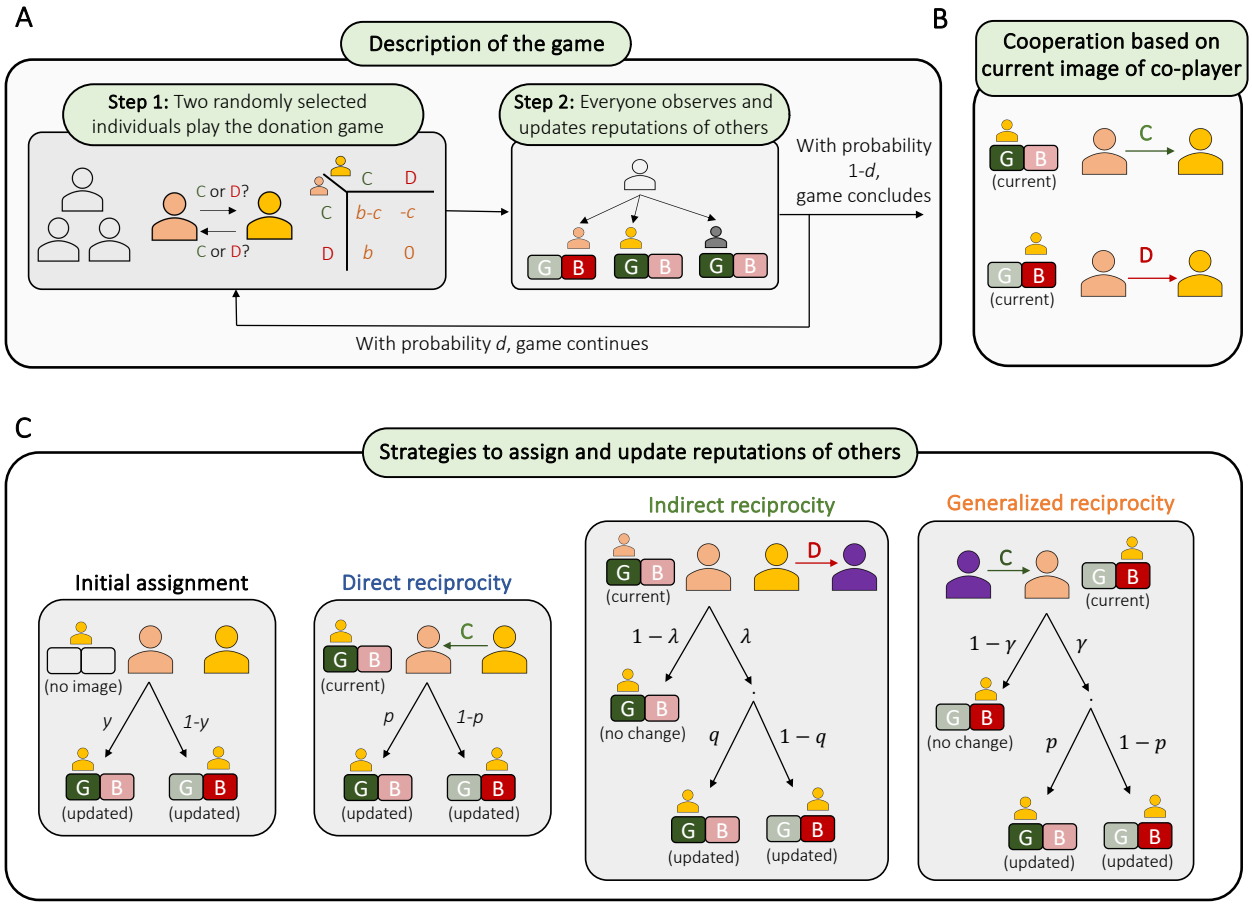


Fig. 1: A model with direct, indirect and generalized reciprocity. **(A)** Individuals in a population of size n interact repeatedly in pairs, where each interaction involves two randomly selected players who independently choose whether to cooperate or defect. Cooperation results in a cost c and a benefit $b > c$ to the co-player, while defection incurs no cost or benefit. Third-party observers, not involved in the interaction, observe the outcome, though these observations may be erroneous with probability ε. Interactions continue with probability d, and the game ends with probability $1 - d$. **(B)** Players assign reputations (good or bad) to their co-players, and their decision to cooperate or defect depends on the current perception of their co-player's standing. **(C)** Players assign a good reputation to co-players for whom there is no existing assessment with a probability y. Reputations are updated based on both direct interactions and indirect information, with parameters p and q governing updates after cooperation or defection, respectively. Indirect reciprocity allows reputation updates from third-party interactions with probability λ, and generalized reciprocity allows spillover updates for uninvolved individuals with probability γ.

observed action within the population. These properties make the model mathematically tractable, allowing for the explicit computation of expected payoffs for strategies in any arbitrary population. To this end, we derive a recursion formula that relates the probability of a focal player considering a given co-player as "good" after the next interaction to the probability that they currently see them

as "good". For a complete description of how payoffs are computed, see Sections 1 and 2 in the **Supplementary Information**.

Nash equilibria and the conditions for cooperation across reciprocities

A strategy is a Nash equilibrium if no player has an incentive to deviate from it, given that the other players also follow it. Our setup allows us to determine whether an arbitrary strategy σ is a Nash equilibrium. The conditions for this are provided in Section 3 of the **Supplementary Information**. Here, we focus on a specific subset of Nash equilibria that give rise to full cooperation, namely, cooperative Nash equilibria. For a strategy to be a cooperative Nash equilibrium, it must be both a Nash equilibrium and self-cooperative. Self-cooperative strategies ensure cooperation in a homogeneous population, given that there are no errors. In our framework, a strategy σ is self-cooperative if and only if $y = p = 1$. We begin by analyzing cooperative equilibria when the population adopts one of the three pure modes of reciprocity.

The existence of cooperative equilibria depends on the model's parameters. For both pure direct and generalized reciprocity, full cooperation is sustainable at equilibrium if, and only if, the expected number of pairwise interactions is sufficiently high. Specifically, the pairwise continuation probability, δ, must satisfy:

$$\delta > \frac{c}{b}. \tag{1}$$

In the case of pure direct reciprocity ($\lambda = \gamma = 0$) with $y = p = 1$, full cooperation is a Nash equilibrium if

$$q \leq q_{\mathrm{D}}^{*} := 1 - \frac{c}{b\delta}. \tag{2}$$

Similarly, for generalized reciprocity ($\lambda = 0, \gamma = 1$), full cooperation is maintained if

$$q \leq q_{\mathrm{G}}^{*} := 1 - \frac{c + c\delta(n-2)}{b\delta + c\delta(n-2)}. \tag{3}$$

From conditions (2) and (3), it can be derived that a fully cooperating population using direct reciprocity can tolerate higher levels of generosity towards errors compared to a population using

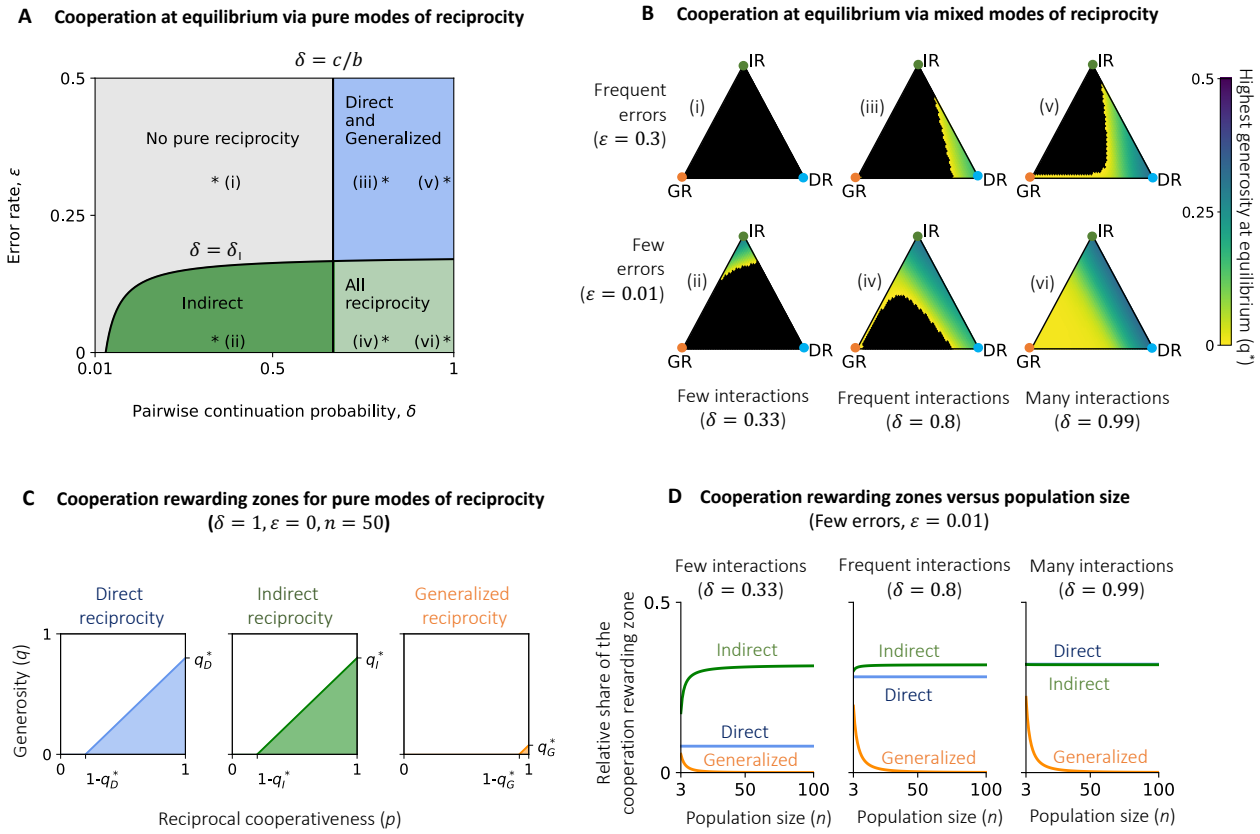


Fig. 2: Equilibrium analysis and cooperation rewarding zones. (A), Pure direct and generalized reciprocity can sustain full cooperation at equilibrium if and only if the expected number of pairwise interactions is sufficiently high (i.e, the pairwise continuation probability, $\delta > c/b$). On the other hand, the condition for indirect reciprocity depends both on the expected number of pairwise interactions and the rate at which observation errors are made: $\delta > \delta_{\mathrm{I}}$; see Eq. (4). **(B),** All possible mixtures of direct, indirect and generalized reciprocity can be represented using a simplex wherein each corner represents pure direct (DR), indirect (IR) and generalized (GR) reciprocity and points inside (or on the edges) represent mixtures of all three (or two) modes of reciprocity. On these simplices, we plot the highest generosity towards defection, q^*, that sustains full cooperation at equilibrium (if full cooperation is feasible). The black regions in the simplices indicate that there is no cooperative Nash equilibrium for the corresponding mixtures of the parameters (see **Materials and Methods** for more details). **(C),** We consider an infinitely long game ($\delta = 1$) where individuals make no errors in observation ($\varepsilon = 0$). The cooperation rewarding zones, for each mode of reciprocity is annotated with colors in the strategy space (p, q). **(D),** We plot the relative size of the cooperation rewarding zone as a fraction of the entire strategy space, for each mode of reciprocity as we vary the population size, n. Unless otherwise stated, we consider a population of 50 individuals. For **A, B** we take benefit to cost ratio, $b/c = 1.5$. For **C, D** we use $b/c = 5$.

generalized reciprocity (i.e., $q^*_{\mathrm{D}} > q^*_{\mathrm{G}}$). Moreover, this difference becomes increasingly significant in larger populations (i.e., $q^*_{\mathrm{D}} \gg q^*_{\mathrm{G}}$ when $n \gg 2$).

Pure indirect reciprocity allows full cooperation at equilibrium even when the expected number

of pairwise interactions is low. However, this requires sufficiently reliable observations of third-party interactions. Specifically, the continuation probability, δ, and the probability of observational errors, ε, must satisfy:

$$\delta > \delta_{\mathrm{I}} := \frac{c}{b + (n-2)((1-2\varepsilon)b - c)}, \tag{4}$$

Additionally, the strategy must satisfy:

$$q \leq q_{\mathrm{I}}^{*} := 1 - \frac{1 + (n-2)\delta}{1 + (n-2)(1-2\varepsilon)} \frac{c}{b\delta}. \tag{5}$$

We also compare the maximum level of generosity that a fully cooperating population can adopt at equilibrium when using indirect reciprocity versus the other modes of reciprocity. When observational errors are sufficiently low, or the benefit-to-cost ratio is sufficiently high (i.e., when $b/c > 1/(1-2\varepsilon)$), a population cooperating via indirect reciprocity can be more generous at equilibrium than one adopting generalized reciprocity (and *vice versa*). Moreover, when observational errors are sufficiently low or the pairwise continuation probability is sufficiently high, a population cooperating via indirect reciprocity can be more generous at equilibrium than one using direct reciprocity. In particular, when $\delta < 1 - 2\varepsilon$.

The differences in the maximum level of generosity that can be sustained at equilibrium across the modes of reciprocity offer insight into how each reciprocity promotes cooperation. To this end, we use the concept of a *cooperation rewarding zone* from the literature on the repeated prisoner's dilemma (35). For a fixed mode of reciprocity, the cooperation rewarding zone refers to the set of all strategies that, when adopted uniformly by a population, make always-cooperate the best response for any individual (see **Materials and Methods** for details). In Fig. 2C, we annotate these sets for all three modes of reciprocity for the infinitely repeated game with no errors ($\delta = 1, \varepsilon = 0$). We observe that the sizes of the cooperation rewarding zones depend on the mode of reciprocity. In particular, for generalized reciprocity, the zone is remarkably small. Furthermore, for generalized reciprocity, the size of the cooperation rewarding zone decreases sharply with an increase in population size (Fig. 2D), indicating that cooperation is less likely to be sustained when large populations adopt generalized reciprocity. In contrast, for direct and indirect reciprocity, the sizes of the cooperation rewarding zones remain relatively constant across different population sizes.

Beyond pure reciprocity modes, our equilibrium analysis predicts cooperative equilibria when

individuals use all three forms of reciprocity simultaneously (see Proposition 1 in the **Supplementary Information**). We perform numerical computations to demonstrate this result in Fig. 2B. We observe that more mixed cooperative Nash equilibria exist when pairwise interactions are longer and observation errors are unlikely. Furthermore, regardless of model parameters, generosity decreases as more generalized reciprocity is used.

The evolution of cooperation when the mode of reciprocity is fixed

The equilibrium analysis has shown that strategies capable of sustaining a cooperative Nash equilibrium exist for all three modes of reciprocity. However, it remains unclear whether these strategies would evolve if individuals can adapt their strategies over time through social learning. To explore this, we conduct evolutionary simulations where individuals update their strategies by imitating others in the population. Additionally, occasional mutations introduce new strategies into the population. We first consider simulations where everyone in the population is restricted to the same mode of reciprocity (i.e., pure direct, indirect, or generalized reciprocity). As a result, players are only able to evolve the strategy components y, p, and q. Furthermore, we assume that mutations are relatively rare, leading to mostly homogeneous populations where all players adopt the same strategy. When a mutant appears, it either becomes fixed in the population (becoming the new resident) or goes extinct. This scenario is known as the limit of rare exploration (or the limit of rare mutation) (36, 37). For a detailed description of the evolutionary process, see **Materials and Methods**.

In simulations where players have very few interactions ($\delta = 0.5$), cooperative Nash strategies are adopted by the population only in the case of indirect reciprocity. For the other two modes of reciprocity, the population predominantly adopts defecting strategies. However, as the number of interactions increases ($\delta = 0.99$), direct reciprocity leads to a predominantly cooperative population, surpassing the cooperation rate of indirect reciprocity. In contrast, generalized reciprocity continues to result in defecting strategies. Thus, consistent with earlier models (14, 32), we find that generalized reciprocity alone is insufficient to produce cooperation in large populations. Further simulations (summarized in Fig. S2) confirm that generalized reciprocity is unable to stabilize cooperation across varying population sizes, selection strengths, and benefit-to-cost ratios.

To further clarify these results, we compare the evolutionary robustness of cooperative and defecting strategies for each type of reciprocity. We focus on two strategies: one that almost always reciprocates cooperation (CC) and one that almost always defects (ALLD). Specifically, the ALLD strategy is defined as $(0.01, 0.01, 0.01, \lambda, \gamma)$, while the CC strategy is defined as $(0.99, 0.99, 0.5, \lambda, \gamma)$. We consider a homogeneous resident population that adopts either the CC or ALLD strategy. We

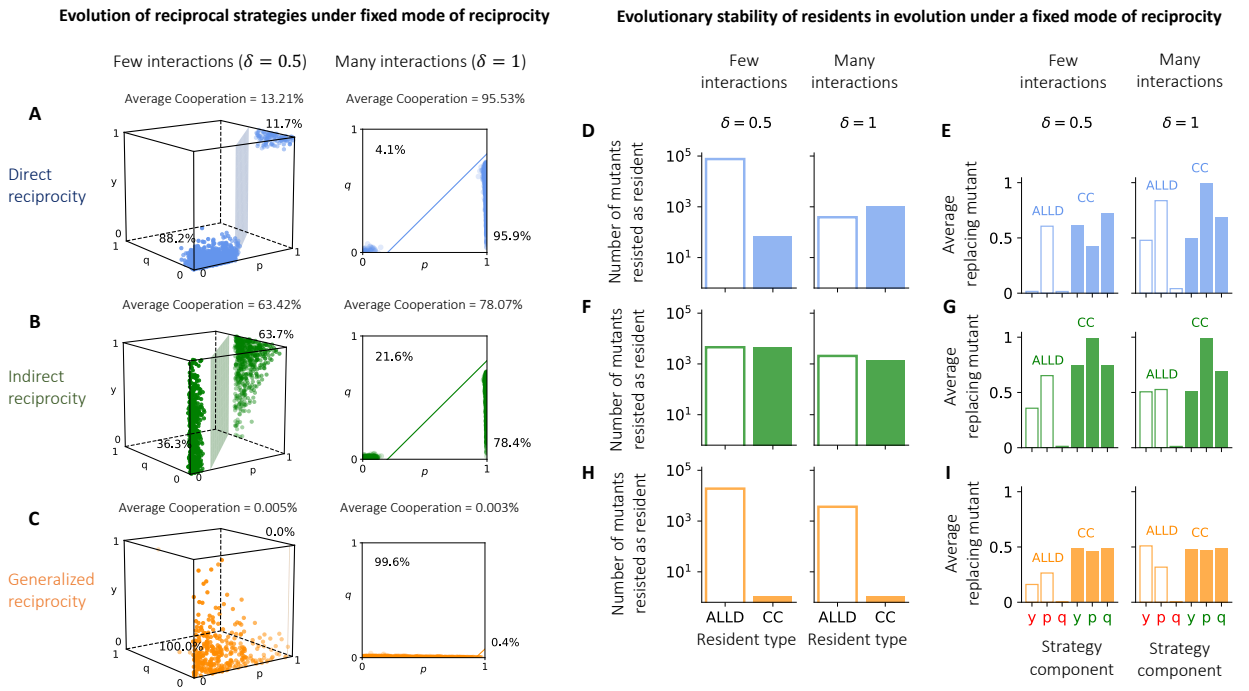


Fig. 3: Evolutionary dynamics at rare exploration under a single mode of pure reciprocity. We perform evolutionary simulations in the limit of rare exploration wherein individuals were restricted to use either pure direct reciprocity (A, D, E), pure indirect reciprocity (B, F, G) or pure generalized reciprocity (C, H, I). In the evolutionary process, individuals were free to choose the strategy elements y, p and q from the unit cube. **(A), (B), (C):** In these plots, the dots indicate the most successful residents in terms of resisting invasion from mutants. In the case of direct and indirect reciprocity, the most successful residents form two major clusters in the strategy space - they are either in the vicinity of the strategy that always defects, $y \approx p \approx 0$, or in the vicinity of the strategy that cooperates conditionally: $p \approx 1$. While using generalized reciprocity, successful strategies almost always defect $y \approx q \approx 0$. The percentages represent the time spent by the whole evolutionary process on each side of the solid plane (or line) which contains all equalizer strategies. At $\delta = 1$, the effect of cooperation in the first round y, is negligible and therefore the strategy space is sufficiently represented by a square instead of a cube. **(D), (F), (H):** For theses plots, we conduct simulations to study the evolutionary robustness of cooperation and defection under each mode of reciprocity (for the exact details of the strategies ALLD type and CC type see **Materials and Methods**). Effectively, we measure time with the number of mutants the residents resist before being invaded by a mutant. **(E), (G), (I):** In here, we plot the strategy components of average mutant that successfully invade the ALLD type and CC type residents in evolutionary simulations. Parameters: $n = 50, b = 5, c = 1, \beta = 10, \varepsilon = 0$ and $\delta \in \{0.5, 1\}$.

then simulate the evolutionary process by repeatedly introducing random mutant strategies and record how many mutants, on average, are needed until one successfully replaces the resident. Additionally, we record the properties of the mutant strategies that successfully invade these populations. The results are summarized in Fig. 3D-I.

For indirect reciprocity, the robustness of cooperative and defective strategies is similar, regardless of the number of interactions. ALLD is always replaced by more cooperative mutants, while CC is replaced by Generous TFT like strategies. In contrast, for direct reciprocity, the CC strategy is more stable than ALLD when the number of interactions is high, but the opposite is true when the number of interactions is low. This explains why the population adopts defecting strategies for small δ. When players rely solely on generalized reciprocity, the CC strategy is highly unstable, regardless of game length. In both short and long games, CC is quickly replaced by almost any mutant due to its excessive generosity toward defectors (Fig. 3I). In contrast, ALLD remains stable under generalized reciprocity.

The co-evolution of direct, indirect and generalized reciprocity

The analysis in the previous section highlights when cooperation is likely to evolve when individuals rely on a single mode of reciprocity. However, reciprocal strategies themselves are subject to selection. We now explore a more general scenario where individuals can evolve their strategy elements y, p, and q, while also choosing between one of the three pure modes of reciprocity. We perform evolutionary simulations in the limit of low mutations and compare the outcomes of our unified model with those of Schmid et al. to understand the impact of introducing generalized reciprocity. We refer to Schmid's model as the "reduced model" and ours as the "extended model". In the reduced model, the population primarily relies on indirect reciprocity, except when observation errors are high, making third-party observations less reliable. In the extended model, however, the population predominantly adopts generalized reciprocity, which leads to lower cooperation rates. Across most of the parameter space, by varying δ (the expected number of interactions) and ε (the probability of errors), we observe a higher cooperation rate in the reduced model in comparison to the extended model.

To understand why the inclusion of generalized reciprocity undermines the evolution of cooperation in the limit where individuals rarely explore, we closely examine one particular simulation (i.e., where $\delta = 0.999$ and $\varepsilon = 10^{-3}$). For these parameters, we observe that while in the reduced setup cooperation is abundant via direct reciprocity, in the extended setup, individuals adopt direct (38%), indirect (32%), and generalized (30%) reciprocity equally often (see Fig. S4A). In particular, they cooperate rarely when adopting generalized reciprocity. By studying the evolu-

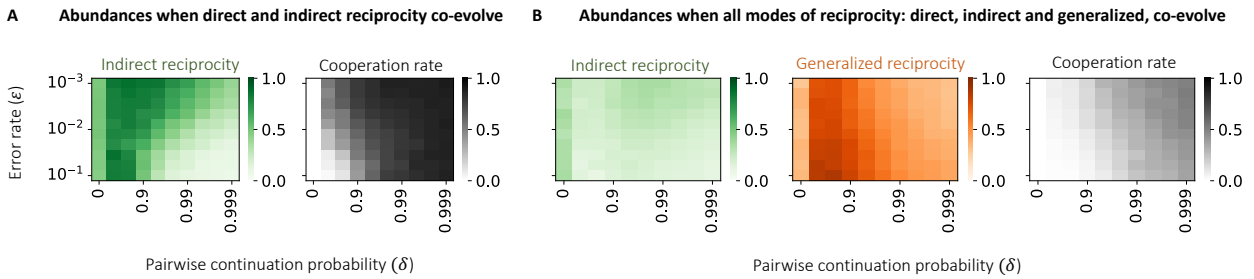


Fig. 4: A comparison between evolutionary simulations with and without generalized reciprocity demonstrates its effect on the evolution of cooperation. In the evolutionary simulations that we conduct for this figure, individuals can choose to adopt different modes of reciprocity. In **(A)**, individuals are restricted to adopt either direct reciprocity or indirect reciprocity while in **(B)**, individuals can choose between the three modes of reciprocity. The evolutionary process is conducted for rare exploration. See **Materials and Methods** for more details on the evolutionary simulations. We present the time average of the abundance of each mode of reciprocity and report how abundant cooperation is in the evolving population. We present these averages as we vary the parameters of the game, the pairwise continuation probability δ and the observation error rate, ε. All other parameters of the game and the evolutionary process are the same as in Fig. 3

tionary transitions in this example, we identify that the key mechanism behind the breakdown of cooperation in the extended setup is neutral selection (see Fig. S4B). In the extended setup, populations that cooperate via direct reciprocity are neutrally invaded by mutants that also cooperate strongly, but via generalized reciprocity. After invasion, these cooperative pay-it-forward residents are themselves susceptible to invasion by defectors. Once defection becomes prevalent, generalized reciprocity is strongly favored. Afterward, transitions to any mode of reciprocal cooperation are unlikely.

Co-evolution of reciprocities when individuals frequently explore new strategies

In the limit where individuals rarely explore new strategies, our analysis predicts weak reciprocal cooperation. To study the role that exploration plays in the evolution of reciprocal cooperation, we perform evolutionary simulations in which individuals randomly adopt arbitrary strategies at a fixed rate, μ (Fig. 5). In these simulations, we study outcomes for long games ($\delta = 0.999$). Unlike simulations for rare exploration, here populations are often composed of more than two strategies. For a detailed description of this evolutionary process see **Materials and Methods**.

Our analysis identifies that an intermediate rate exploration has a positive effect on the evolution of cooperation, even in the presence of generalized reciprocity. When the rate of exploration is neither too high nor too low, individuals strictly prefer to cooperate via direct reciprocity. In contrast,

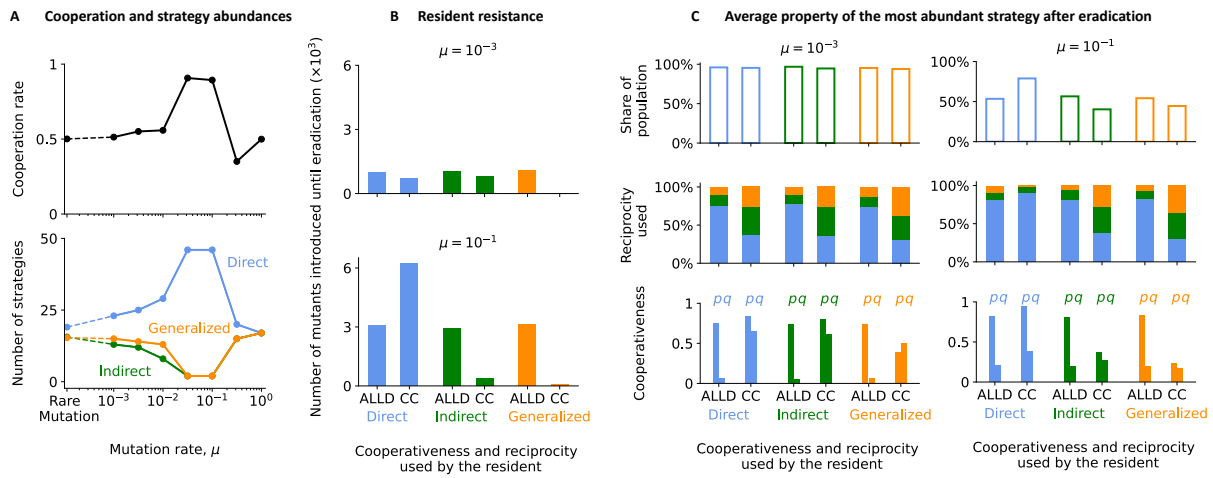


Fig. 5: The co-evolution of reciprocal strategies, when exploration is frequent, leads to the emergence of stable and abundant cooperation. We conduct evolutionary simulations in which individuals frequently adopt random strategies thereby leading to mixed populations. **(A),** For each simulation we perform, we fix the rate at which individuals adopt a random strategy, μ (also called the mutation or the exploration rate). From each simulation, we report the average cooperation rate and the average composition of the population, with respect to the reciprocal strategies that individuals adopt. At intermediate values of μ we observe that cooperative individuals who adopt direct reciprocity dominate the evolving population. **(B),** We conduct an invasion experiment to study how resistant reciprocal strategies are to the process of mutation and selection. To this end, we take six different resident populations, two for each mode of reciprocity – ALLD and CC (see **Materials and Methods** for exact details) and report the average number of mutations necessary to completely eradicate the incumbent population. We conduct these simulations for two values of μ: 10^{-3} and 10^{-1}. When exploration is intermediate ($\mu = 10^{-1}$), the cooperative direct reciprocity population displays strong resistance. **(C)**, From the simulations we perform to study the invasion resistance, we report the average property of the most abundant strategy after the eradication is complete. We report the share of the population that the most abundant strategy occupies on average (top row), the estimated likelihood of the reciprocal strategy that it uses (middle row) and the average values of p and q of its strategy (bottom row). Parameters: $\delta = 0.999$, $\varepsilon = 0$, b, c and β are kept the same from Fig. 3

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when rates of exploration are either too high or too low, individuals try all modes of reciprocity about equally but cooperation rates drop to approximately 50% (Fig. 5A).

We perform additional simulations to examine the reason why cooperation via direct reciprocity dominates at intermediate rates of exploration (Fig. 5B,C). We conduct evolutionary simulations starting with a unique homogenous resident population each time and study (i) the average time it takes to eradicate the resident strategy completely and (ii) the average property of the most abundant strategy after the eradication is complete. We observe that when individuals explore at an intermediate rate ($\mu = 10^{-1}$), populations which use direct reciprocity and cooperate highly take the longest to eradicate (on average). Additionally, when they are indeed eradicated, the result-

ing population is usually abundant with individuals who also use direct reciprocity to cooperate. In contrast, other resident populations are relatively less stable to this process of exploration and learning. Moreover, if the initial resident population cooperates lowly, they are usually replaced by a population that is abundant with individuals who cooperate via direct reciprocity. At intermediate rates of exploration, cooperation through direct reciprocity exhibits greater stability and is more prone to positive selection when contrasted with other kinds of reciprocal behaviour.

DISCUSSION

In repeated interactions, humans often use past outcomes to guide current decisions (3, 27, 38–40). They may recall how others have treated them (generalized reciprocity), their own past interactions with the same person (direct reciprocity), or their knowledge of how the person treated others (indirect reciprocity). Despite the different ways humans reciprocate, it remains unclear when one form of reciprocity is preferred over another. A recent study (28) explored this question using a model limited to direct and indirect reciprocity. Here, we extend this work by including generalized reciprocity. We derive a method to compute payoffs when players combine all three strategies and characterize the resulting Nash equilibria, providing explicit conditions for each form of reciprocity to sustain cooperation.

We find that generalized reciprocity alone can stabilize cooperation in populations expecting frequent interactions. However, generalized reciprocity often undermines cooperation when players imitate others, leading to widespread defection (Fig. 4). Evolutionary simulations show that populations reliant on generalized reciprocity are vulnerable to defection, ultimately collapsing into non-cooperative behavior (Fig. S4). Stable cooperation is only maintained when players explore new strategies at a steady rate. This observation resonates with a recent finding which also reports a strong correspondence between exploration and cooperation via direct reciprocity (41).

Our results on the limits of generalized reciprocity (Fig. 3C) agree with previous models (14, 31, 32, 42). Cooperation through generalized reciprocity alone rarely emerges in random interactions without additional mechanisms like assortment (31), group movement (42), or immediate reciprocal payback (32).

However, our view on the emergence of generalized reciprocity diverges from earlier work. While prior models explain why cooperation can evolve through generalized reciprocity with supporting mechanisms, they do not explain why generalized reciprocity evolves in the first place. Our results show that when competing forms of reciprocity are allowed, generalized reciprocity is rarely favored (Figs. 4, 5). Even if additional mechanisms favor it, it is unclear why direct or

indirect reciprocity wouldn't be equally selected, as seen in network-based models of direct reciprocity (43, 44).

Nowak and Roch (32) suggest that direct reciprocity can promote generalized reciprocity, but our results show the opposite. Players tend to avoid generalized reciprocity, favoring defection when direct and generalized reciprocity compete (Fig. S5). We attribute this discrepancy to differences in how direct reciprocity is modeled. In our study, individuals choose between their general experience and specific interactions, whereas in Nowak and Roch's model, direct reciprocity only occurs after receiving cooperation, functioning more like a reward system (45).

Sasaki et al. (33) propose a strategy that simultaneously applies generalized and indirect reciprocity while interacting in an infinitely large population. When a player uses this strategy, they invariably cooperate after receiving a cooperation. In case they do not receive a cooperation, they cooperate based on the reputation of their beneficiary (which could be assigned with a simple scoring rule). The authors refer to this conditional strategy as integrated reciprocity. They observe that integrated reciprocity is able to stably coexist with unconditional defectors in games that are infinitely long. Even though defectors take advantage of individuals who use integrated reciprocity, the strategy does not die out because rewards for being good' (from other integrated reciprocators) balance out the exploitation. This result indicates that a coexistence between cooperative generalized reciprocity and cooperative indirect reciprocity may stably emerge in our model too. However, we do not detect any such stable coexistence in our evolutionary simulations (Fig. 5A). We argue that this is because players can sometimes adopt random strategies in our evolutionary process, and more importantly, these strategies can utilise direct reciprocity (a strategy not included in Sasaki et al.). When games are long, cooperation via direct reciprocity is very likely to spread in any population that uses a combination of generalized and indirect reciprocity.

Generalized reciprocity is often argued to evolve due to its lower cognitive demands (14–16), compared to direct or indirect reciprocity (46–48). We do not account for cognitive costs in our model. However, we note that it is plausible that once (in)direct reciprocity is considered costly in this model, the emergence of defection via generalized reciprocity becomes even more likely. Therefore, it is still not clear how exactly individuals with limited memory would learn to cooperate.

Lastly, we address the apparent conflict between our findings and empirical studies that show humans engage in generalized reciprocity (7–9, 25, 26). We propose that pay-it-forward cooperation may have evolved independently of direct or indirect reciprocity. Our model serves as a counterfactual where all forms of reciprocity co-evolve to maximize payoffs. In this scenario, we would expect minimal reciprocal cooperation or generalized reciprocity to be undermined by direct

reciprocity, as observed. This suggests that generalized reciprocity may have emerged via mechanisms beyond material payoffs. Supporting this view, a neurological study (49) shows that pay-it-forward cooperation is driven by empathy, while reputation-based reciprocity engages self-centered cognition, hinting that generalized reciprocity may evolve through non-material motivations.

MATERIALS AND METHODS

Cooperation rewarding zone

We consider that a population of size n individuals where everyone uses the strategy $\sigma := (y, p, q, \lambda, \gamma)$. Then, the resident strategy σ is said to belong in the cooperation rewarding zone if a single individual can best-respond by unilaterally switching to a strategy that cooperates in every round. If a single individual's best-response is to switch to always-defect, then σ is said to belong in the defection rewarding zone. If σ belongs in neither the cooperation nor the defection rewarding zone, it belongs in the set of equalizer strategies. In this case, any strategy is a best-response for a single unilaterally deviating individual. These three sets are disjoint and their union is the entire strategy space, $[0, 1]^5$. If the resident population uses pure direct, pure indirect or pure generalized reciprocity, the cooperation rewarding zone is the subset of the strategy space in which the strategy components p and q satisfy $p - q > 1 - q^*$. Here q^* is either q^*_{D}, q^*_{I} or q^*_{G}, from Eqs. (2), (5) and (3) respectively, depending on the mode of reciprocity employed by the resident population. If $p - q < 1 - q^*$, the strategy belongs in the defection-rewarding zone. If $p - q = 1 - q^*$, the strategy is an equalizer strategy.

Evolutionary dynamics at any arbitrary mutation rate

At each time step of our evolutionary process, a focal individual is picked uniformly at random from the population to update their strategy. With probability μ, the focal player picks a strategy uniformly at random from the set of permissible strategies, and adopts it. With probability $1 - \mu$, the focal player picks another individual from the population, uniformly at randon (the role model), and adopts their current strategy with probability

$$\rho_{\mathrm{F}\to\mathrm{R}} = \frac{1}{1 + e^{\beta(\pi_{\mathrm{F}} - \pi_{\mathrm{R}})}}. \tag{6}$$

Here π_{F} and π_{R} are the expected payoffs of the focal and the role model individuals in the current population. The parameter β is called the selection strength. It reflects the importance that

players assign to payoff differences while imitating a new strategy. The computation of expected payoffs of individual strategies at any arbitrary population composition is discussed in **Supplementary Information**.

Evolutionary dynamics at rare exploration

The evolutionary process in the limit of rare exploration ($\mu \to 0$) assumes that a single mutant in a homogenous population either undergoes fixation or goes extinct before the next mutant arises. In other words, the expected time-interval between the arrival of two mutants is sufficiently larger than the expected time of fixation (or extinction) of a single mutant. This assumption largely simplifies the evolutionary process. With this, one can assume that the population is almost always composed by a single strategy. The entire history of the evolutionary process can be summarized by recording a) the sequence of resident strategies that occupied the population and b) the length of evolutionary time that each resident strategy occupied the population (in other words, the number of mutants they resisted an invasion from). The outcome of each simulation is thus a sequence $\{\sigma_1, \sigma_2, \sigma_3, ..., \sigma_l, ...\}$ of resident strategies, together with a list $\{t_1, t_2, t_3, ..., t_l, ...\}$ that denotes how long each resident was present.

For the mutant fixation dynamics, we use the imitation process described in the previous section. In this dynamical process, at every time step, a pair of focal and role model players are picked uniformly at random from the population. The focal player compares their current expected payoff π_{F}, with the current expected payoff of the role model, π_{R}, and switches to the role model's strategy with probability, $\rho_{\mathrm{F}\to\mathrm{R}}$ given in Eq. (6). The probability that a single mutant σ_M, attains fixation when introduced in a homogenous σ_R-playing population is computed exactly using the formula derived in Traulsen et al. (50),

$$f(\sigma_M, \sigma_R) = \frac{1}{1 + \sum_{m=1}^{n-1} \prod_{k=1}^{m} e^{-\beta(\pi_M(k) - \pi_R(k))}}. \tag{7}$$

Here, $\pi_M(k)$ and $\pi_R(k)$ are respectively the expected payoffs of the mutant and the resident strategy when k individuals play σ_M and $n-k$ individuals play σ_R. We compute this probability in our simulations to determine whether a newly introduced mutant takes over the population.

Methods for figures

Methods for Fig. 2, In Fig 2A, we consider a population of 50 individuals and plot the regions in the δ-ε plane corresponding to regions where strategies employing pure direct, pure indirect and pure generalized reciprocity are able to constitute a cooperative Nash equilibria. The regions are marked by the two boundaries: $\delta = \delta_{\mathrm{D}} = \delta_{\mathrm{G}}$ and $\delta = \delta_{\mathrm{I}}$. The expressions for δ_{D}, δ_{I} and δ_{G} are given in Eq. (1) and Eq. (4).

In Fig 2B, we use the simplex representation of reciprocal strategies to denote whether cooperative strategies of the form $(1, 1, q, \lambda, \gamma)$ are able to constitute a Nash equilibria. Each point inside the triangular simplex $\mathcal{S}_{\mathrm{DIG}}(n)$ represents a tuple of three complimentary probabilities $(\alpha_{\mathrm{D}}, \alpha_{\mathrm{I}}, \alpha_{\mathrm{G}})$ and corresponds to a unique pair $(\lambda, \gamma) \in [0, 1]^2$. The probabilities, $\alpha_{\mathrm{D}}, \alpha_{\mathrm{I}}$ and α_{G} represent the effective likelihood with which individuals in a homogenous population use direct, indirect and generalized reciprocity respectively. For a precise mathematical definition of these probabilities, see the section on 'effective likelihood of reciprocity' in **Supplementary Information**. The three corners of the triangular simplex correspond to three modes of pure reciprocity. Pure direct ($\lambda = 0$, $\gamma = 0$), pure indirect ($\lambda = 1$, $\gamma = 0$) and pure generalized ($\lambda = 0$, $\gamma = 1$) reciprocity correspond to the corners of the simplex: $(1, 0, 0)$, $(\alpha_{\mathrm{D}}^{min}, \alpha_{\mathrm{I}}^{max}, 0)$ and $(\alpha_{\mathrm{D}}^{min}, 0, \alpha_{\mathrm{G}}^{max})$ respectively. These corners are marked by DR, IR and GR in Fig. 2B. The relative distance of a point from the corners of the simplex represents how often individuals use different kinds of reciprocity to update reputations in the corresponding homogenous population. For several discrete points inside the simplex, we numerically check whether there is a $q \in [0, 1)$ such that $(1, 1, q, \lambda, \gamma)$ is a Nash equilibrium strategy. The black regions on the simplices correspond to populations which cannot constitute a fully-cooperative Nash equilibria. When cooperative Nash equilibria exist, we plot the highest value of q (generosity towards defection), q^*, that sustains full cooperation in equilibrium.

In Fig 2C we consider a a game with $n = 50$ players that is infinitely long ($\delta = 1$) and in which players do not make any observational errors ($\varepsilon = 0$). For each mode of reciprocity, we annotate regions in the strategy space (p, q) in which every strategy rewards cooperation (or equivalently, falls in the cooperation rewarding zone. See the section on cooperation rewarding zone for more details). The cooperation rewarding zone for a fixed mode of reciprocity is a triangle subtended by the points $(1 - q^*, 0), (1, q^*)$ and $(1, 0)$ in the strategy space (p, q) where q^* is q_{D}^*, q_{I}^* or q_{G}^* depending on the mode of reciprocity. In Fig 2D we plot the sizes of these triangles, relative to the whole strategy space, for a game in which players make occasional errors ($\varepsilon = 0.01$), as we vary the population size n.

Methods for Fig. 3, For this figure, we conduct evolutionary simulations at rare exploration (mutations). For panels A, D and E, we constrain the permissible set of strategies to $\lambda = \gamma = 0$

(pure direct reciprocity). Similarly, for panels B, F and G we constrain it to $\lambda = 1, \gamma = 0$ (pure indirect reciprocity) and for panels C, H and I to $\lambda = 0, \gamma = 1$ (pure generalized reciprocity). In the simulations for panels A, B, C, 10^7 mutants attempted to invade resident populations over the course of the resident replacement process. The simulations began with a random strategy $(y, p, q, \lambda, \gamma)$ where (y, p, q) was uniform randomly drawn from the set $[0, 1]^3$ and λ, γ were set to either $(0, 0)$, $(1, 0)$ or $(0, 1)$ depending on the mode of reciprocity being studied. Subsequent mutants were drawn analogously. Over the course of a single simulation, we kept the parameters fixed at $\varepsilon = 0, c = 1, b = 5, \beta = 10$ and $\delta \in \{0.5, 1\}$.

We use colored dots to denote the 1000 residents that were most successful in resisting invasions from mutants. For the simulation at $\delta = 0.5$ in panel C, we show only 329 (and not 1000) residents, because the process only generated 329 different residents. In every plot in panels A, B and C, the solid plane (or the solid line) in the strategy space (y, p, q) denotes the subspace that contains the equalizer strategies (51). As residents, the equalizer strategies, set the payoff of any arbitrary mutant to a fixed value. As a result, these strategies also constitute Nash equilibria (see Theorem 1 in **Supplementary Information**). These strategies satisfy $p - q = 1 - q^*$. The value of q^* depends on the mode of pure reciprocity: it is q_D^* for pure direct reciprocity, q_I^* for pure indirect reciprocity and q_G^* for pure generalized reciprocity. See Eqs. (2), (5) and, (3).

For panels D-I, we conduct multiple independent evolutionary simulations that begin with either an ALLD type or a CC type strategy. Here, the ALLD type strategy is $(0.01, 0.01, 0.01, \lambda, \gamma)$ while the CC type is $(0.99, 0.99, 0.5, \lambda, \gamma)$. Each simulation stops when the initial population is taken over by a mutant. We record the number of mutants the initial population resisted before they are invaded. The average of these numbers across 1000 independent simulations are plotted in panels d,f and h. We also record the strategy that successfully invades these two residents in every evolutionary simulation. The average strategy that successfully invades, across 1000 independent simulations, are shown in panels e,g and i.

Methods for Fig. 4, In this figure we compare results from two types of evolutionary simulations. In the first type, strategies that individuals can adopt are restricted to pure direct and pure indirect reciprocity (panel a). That is, they adopt from the set

$$\Sigma_{\mathrm{DI}} := \Big\{ (y, p, q, \lambda, 0) \;\Big|\; (y, p, q) \in [0, 1]^3 \wedge \lambda \in \{0, 1\} \Big\}. \tag{8}$$

In the second type, they are free to choose between all the three types of reciprocity (panels b). That is, they choose from,

$$\Sigma_{\mathrm{DIG}} := \Big\{ (y, p, q, \lambda, \gamma) \;\Big|\; (y, p, q) \in [0, 1]^3 \wedge (\lambda, \gamma) \in \{(0, 0), (1, 0), (0, 1)\} \Big\} \tag{9}$$

All simulations are conducted in the limit of rare exploration. The parameters δ, ε, b, c and β remain fixed over the course of a single simulation. Each simulation is ran until 10^7 mutants. We calculate average properties of the resident strategies that occupy the population in these simulations. We plot the time average of the likelihood that individuals use indirect reciprocity and generalized reciprocity to update reputations (for more details on these computations see the section on effective likelihood of reciprocity in **Supplementary Information**). We also compute the likelihood that individuals cooperate in pairwise interactions. The cooperation rate in each homogenous population is calculated using the result in Proposition 1 in **Supplementary Information**. In Fig. S3, we plot the average time series of these likelihoods and cooperation rate for specific values of the parameters δ and ε.

Methods for Fig. 5, For this figure we conduct evolutionary simulations at arbitrary mutation rates. For all simulations we perform, we fix the set of permissible strategies Σ to $\Sigma_{\rm DIG}$ as given by Eq. (9). All simulations begin with a homogenous population in which everyone adopts the strategy that defects in every round $(0, 0, 0, \lambda, \gamma)$. For figures in panel **A**, we ran our simulations for 10^7 time steps when $\mu \in \{10^-3, 10^{-2.5}\}$ and for 10^6 times steps when $\mu \in \{10^{-2}, 10^{-1.5}, 10^{-1}, 10^{-0.5}, 1\}$. For each simulation we conduct, we report the expected frequency of cooperative interactions, averaged over all populations that emerge in the process. We also report the relative frequency of the three modes of reciprocity, averaged over entire simulations. For figures in panel **B**, we still keep $\Sigma = \Sigma_{\rm DIG}$ but start with different initial populations. We study six initial populations, two for each mode of reciprocity: the ALLD and the CC type (similar to Fig. 3). Here the ALLD type is a strategy of the form $(0.01, 0.01, 0.01, \lambda, \gamma)$ while the CC type is $(0.99, 0.99, 0.4, \lambda, \gamma)$. For each initial population, we conduct 1000 independent simulations. We continue these simulations until the first time all individuals adopt a different strategy than the one they assumed at the beginning of the simulation. We estimate the time it takes for this event to occur by averaging over the independent simulations. Across simulations, we also compute average composition of the population when this event occurs.

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Data and materials availability. All code used to carry out the evolutionary simulations is available in an online repository, and the data used to produce the manuscript figures have been archived. More details can be found in (52).

References

[1] M. A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006.

[2] C. Wedekind and M. Milinski. Cooperation through image scoring in humans. *Science*, 288:850–852, 2000.

[3] M. Milinski, D. Semmann, T. C. Bakker, and H.-J. Krambeck. Cooperation through indirect reciprocity: image scoring or standing strategy? *Proceedings of the Royal Society B*, 268:2495–2501, 2001.

[4] I. Seinen and A. Schram. Social status and group norms: Indirect reciprocity in a repeated helping experiment. *European Economic Review*, 50:581–602, 2006.

[5] D. Engelmann and U. Fischbacher. Indirect reciprocity and strategic reputation building in an experimental helping game. *Games and Economic Behavior*, 67:399–407, 2009.

[6] E. Yoeli, M. Hoffman, D. G. Rand, and M. A. Nowak. Powering up with indirect reciprocity in a large-scale field experiment. *Proceedings of the National Academy of Sciences*, 110(Supplement 2):10424–10429, 2013.

[7] W. E. Baker and N. Bulkley. Paying it forward vs. rewarding reputation: Mechanisms of generalized reciprocity. *Organization Science*, 25(5):1493–1510, 2014.

[8] D. Melamed, B. Simpson, and J. Abernathy. The robustness of reciprocity: Experimental evidence that each form of reciprocity is robust to the presence of other forms of reciprocity. *Science Advances*, 6(23):eaba0504, 2020.

[9] B. Simpson, A. Harrell, D. Melamed, N. Heiserman, and D. V. Negraia. The roots of reciprocity: Gratitude and reputation in generalized exchange systems. *American Sociological Review*, 83(1):88–110, 2018.

[10] M. A. Nowak and K. Sigmund. Evolution of indirect reciprocity. *Nature*, 437:1291–1298, 2005.

[11] H. Ohtsuki and Y. Iwasa. The leading eight: Social norms that can maintain cooperation by indirect reciprocity. *Journal of Theoretical Biology*, 239:435–444, 2006.

[12] H. Ohtsuki and Y. Iwasa. How should we define goodness? – Reputation dynamics in indirect reciprocity. *Journal of Theoretical Biology*, 231:107–20, 2004.

[13] D. Clark, D. Fudenberg, and A. Wolitzky. Indirect reciprocity with simple records. *Proceedings of the National Academy of Sciences*, 117(21):11344–11349, 2020.

[14] T. Pfeiffer, C. Rutte, T. Killingback, M. Taborsky, and S. Bonhoeffer. Evolution of cooperation by generalized reciprocity. *Proceedings of the Royal Society B*, 272:1115–1120, 2005.

[15] Z. Barta, J. M. McNamara, D. B. Huszar, and M. Taborsky. Cooperation among non-relatives evolves by state-dependent generalized reciprocity. *Proceedings of the Royal Society B: Biological Sciences*, 278(1707):843–848, 2011.

[16] C. Rutte and M. Taborsky. Generalized reciprocity in rats. *PLoS Biology*, 5:e196, 2007.

[17] R. Axelrod and W. D. Hamilton. The evolution of cooperation. *Science*, 211:1390–1396, 1981.

[18] G. Van Doorn and M. Taborsky. The evolution of generalized reciprocity on social interaction networks. *Evolution*, 66(3):651–664, 2012.

[19] D. Fudenberg, D. G. Rand, and A. Dreber. Slow to anger and fast to forgive: Cooperation in an uncertain world. *American Economic Review*, 102(2):720–749, 2012.

[20] M. van Veelen, J. García, D. G. Rand, and M. A. Nowak. Direct reciprocity in structured populations. *Proceedings of the National Academy of Sciences USA*, 109:9929–9934, 2012.

[21] C. Hilbe, T. Röhl, and M. Milinski. Extortion subdues human players but is finally punished in the prisoner's dilemma. *Nature Communications*, 5:3976, 2014.

[22] C. Hilbe, K. Chatterjee, and M. A. Nowak. Partners and rivals in direct reciprocity. *Nature Human Behaviour*, 2(7):469–477, 2018.

[23] R. Boyd and P. J. Richerson. The evolution of indirect reciprocity. *Social Networks*, 11(3):213–236, 1989.

[24] M. Dufwenberg, U. Gneezy, W. Güth, and E. Van Damme. Direct vs indirect reciprocity: an experiment. *Homo Oeconomicus*, 18(1/2):19–30, 2001.

[25] K. Herne, O. Lappalainen, and E. Kestilä-Kekkonen. Experimental comparison of direct, general, and indirect reciprocity. *The Journal of Socio-Economics*, 45:38–46, 2013.

[26] L. Stanca. Measuring indirect reciprocity: Whose back do we scratch? *Journal of Economic Psychology*, 30(2):190–202, 2009.

[27] L. Molleman, E. van den Broek, and M. Egas. Personal experience and reputation interact in human decisions to help reciprocally. *Proceedings of the Royal Society B*, 28:20123044, 2013.

[28] L. Schmid, K. Chatterjee, C. Hilbe, and M.A. Nowak. A unified framework of direct and indirect reciprocity. *Nature Human Behaviour*, 5:1292–1302, 2021.

[29] M. Nakamaru and M. Kawata. Evolution of rumours that discriminate lying defectors. *Evolutionary Ecology Research*, 6(2):261–283, 2004.

[30] M. Seki and M. Nakamaru. A model for gossip-mediated evolution of altruism with various types of false information by speakers and assessment by listeners. *Journal of Theoretical Biology*, 407:90–105, 2016.

[31] D. J. Rankin and M. Taborsky. Assortment and the evolution of generalized reciprocity. *Evolution*, 63(7):1913–1922, 2009.

[32] M. A. Nowak and S. Roch. Upstream reciprocity and the evolution of gratitude. *Proceedings of the Royal society B: Biological Sciences*, 274(1610):605–610, 2007.

[33] T. Sasaki, S. Uchida, I. Okada, and H. Yamamoto. The evolution of cooperation and diversity by integrated indirect reciprocity. *arXiv preprint arXiv:2303.04467*, 2023.

[34] U. Berger. Learning to cooperate via indirect reciprocity. *Games and Economic Behavior*, 72:30–37, 2011.

[35] K. Sigmund. *The Calculus of Selfishness*. Princeton Univ. Press, Princeton, NJ, 2010.

[36] L. A. Imhof and M. A. Nowak. Stochastic evolutionary dynamics of direct reciprocity. *Proceedings of the Royal Society B*, 277:463–468, 2010.

[37] S. K. Baek, H-C. Jeong, C. Hilbe, and M. A Nowak. Comparing reactive and memory-one strategies of direct reciprocity. *Scientific Reports*, 6(1):25676, 2016.

[38] B. Rockenbach and M. Milinski. The efficient interaction of indirect reciprocity and costly punishment. *Nature*, 444(7120):718–723, 2006.

[39] I. Okada, H. Yamamoto, Y. Sato, S. Uchida, and T. Sasaki. Experimental evidence of selective inattention in reputation-based cooperation. *Scientific Reports*, 8:14813, 2018.

[40] R. D. Sommerfeld, H.-J. Krambeck, D. Semmann, and M. Milinski. Gossip as an alternative for direct observation in games of indirect reciprocity. *Proceedings of the National Academy of Sciences USA*, 104(44):17435–17440, 2007.

[41] Josef Tkadlec, Christian Hilbe, and Martin A Nowak. Mutation enhances cooperation in direct reciprocity. *Proceedings of the National Academy of Sciences*, 120(20):e2221080120, 2023.

[42] I. M. Hamilton and M. Taborsky. Contingent movement and cooperation evolve under generalized reciprocity. *Proceedings of the Royal Society B: Biological Sciences*, 272(1578):2259–2267, 2005.

[43] H. Ohtsuki and M. A. Nowak. Direct reciprocity on graphs. *Journal of Theoretical Biology*, 247(3):462–470, 2007.

[44] J. M. Pacheco, A. Traulsen, H. Ohtsuki, and M. A. Nowak. Repeated games and direct reciprocity under active linking. *Journal of Theoretical Biology*, 250:723–731, 2008.

[45] K. Sigmund, C. Hauert, and M. A. Nowak. Reward and punishment. *Proceedings of the National Academy of Sciences*, 98(19):10757–10762, 2001.

[46] M. Milinski and C. Wedekind. Working memory constrains human cooperation in the prisoner's dilemma. *Proceedings of the National Academy of Sciences USA*, 95:13755–13758, 1998.

[47] S. Duffy and J. Smith. Cognitive load in the multi-player prisoner's dilemma game: Are there brains in games? *Journal of Behavioral and Experimental Economics*, 51:47–56, 2014.

[48] F. P. Santos, F. C. Santos, and J. M. Pacheco. Social norm complexity and past reputations in the evolution of cooperation. *Nature*, 555:242–245, 2018.

[49] T. Watanabe, M. Takezawa, Y. Nakawake, A. Kunimatsu, H. Yamasue, M. Nakamura, Y. Miyashita, and N. Masuda. Two distinct neural mechanisms underlying indirect reciprocity. *Proceedings of the National Academy of Sciences*, 111(11):3990–3995, 2014.

[50] A. Traulsen, M. A. Nowak, and J. M. Pacheco. Stochastic dynamics of invasion and fixation. *Physical Review E*, 74(1):011909, 2006.

[51] W. H. Press and F. D. Dyson. Iterated prisoner's dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109:10409–10413, 2012.

[52] S. Pal, C. Hilbe, and N. E. Glynatsi. [DATA] The co-evolution of direct, indirect and generalized reciprocity, 2024. Available online at: `https://doi.org/10.5281/zenodo.14035109`.

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Supplementary Information

The co-evolution of direct, indirect and generalized reciprocity

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1 Model

Setup of repeated games in a finite population

We consider a finite population of n individuals (called players). In this population, players only have pairwise interactions with each other. At each (discrete) time point, a single pair interacts in the population. After a pairwise interaction, the next interaction in the population occurs with probability d. In an interaction, two players are randomly drawn from the n players to play the donation game. In the donation game, a player can either cooperate, wherein they give a benefit of b to their co-player while incurring a cost, $c < b$, or defect, in which case they do nothing. Their actions are observable to everyone in the population.

In our model, players can make errors in observing actions in interactions in which they are not taking part (we call them 3rd party interactions). Each player independently mistakes cooperation for defection (and *vice versa*) with probability $\varepsilon \in [0, 1/2]$ in the model. Players do not make an observation error if they are part of the interaction.

We define the pairwise continuation probability, δ, as the conditional probability that a pair of players will meet again in the future if they meet in the present. This probability δ, can be derived in terms of the population continuation probability, d and population size, n as[1],

$$\delta = \frac{2d}{2d + n(n-1)(1-d)}. \tag{1}$$

Strategies of players

In the model, players keep track of others' reputation. To every player in the population, everyone other player is in a binary standing – they are either *good* or *bad*. A player cooperates with their co-player in an interaction if and only if they are in a *good* standing (otherwise they defect).

After an interaction, all players have the chance to update reputations of others by using their strategy and the (possibly erroneous) observation from the interaction. The strategy of a player i, σ_i, is a tuple with five elements: $\sigma_i = (y_i, p_i, q_i, \lambda_i, \gamma_i)$. Each element of the strategy is a probability and is therefore in the interval $[0, 1]$. We describe the strategy components below:

1. y_i (initialization): The probability that i assigns a *good* reputation to a player j when a) i has no reputational history of j (i.e., she has never updated j's reputation before)

2. p_i (reciprocal cooperativeness): The probability that i assigns a *good* reputation to an individual if she observes a cooperation by them in direct interaction.

3. q_i (generosity): The probability that i assigns a *good* reputation to an individual if she observes a defection by them in direct interaction.

4. λ_i (imporatance given to third party interactions, or *indirect reciprocity*): The probability that i updates the reputation of player j after an interaction that involved j and k. While updating, i assumes that j plays against her what she observes him playing against k. That is, she uses the probabilities p_i, q_i and the observed action of j against k to update j's reputation.

5. γ_i (generalizing the action of an interaction partner to the whole population or, *generalized reciprocity*): The probability that i updates the reputation of player j after an interaction that involved i and k but not j. After her interaction with k, player i, chooses to update the reputation of all $j \neq k$ in the population independently with probability γ_i. While updating the reputation of a particular j, i assumes that j plays the same action that k played against her in the interaction.

If player i uses $\lambda_i = 0$ in her strategy, she does not update the reputation of others if she, herself, was not part of the respective interaction. In other words, she only update reputations based on what was done to her (and ignore what others do between each other). If she uses $\gamma_i = 0$, she does not update the reputation of others based on what someone else did to her. We assume players always update the reputation of an individual after directly interacting with them.

Recursion of reputations in an arbitrary population composition

We denote with $x_{ij}(t)$ the probability that a player i considers j *good* after t interactions. This probability at time $t+1$, $x_{ij}(t+1)$, can be written using values of x at time t using the following recursive relation,

$$\begin{aligned} x_{ij}(t+1) = {} & (1 - 2\bar{w} + w)x_{ij}(t) \\ & + w(x_{ji}(t)p_i + (1 - x_{ji}(t))q_i) \\ & + (\bar{w} - w)(1 - \lambda_i)x_{ij}(t) \\ & + w\lambda_i \sum_{l \neq i,j} (\bar{\varepsilon}x_{jl}(t) + \varepsilon\bar{x}_{jl}(t))p_i + (\bar{\varepsilon}\bar{x}_{jl}(t) + \varepsilon x_{jl}(t))q_i \\ & + (\bar{w} - w)(1 - \gamma_i)x_{ij}(t) \\ & + w\gamma_i \sum_{l \neq i,j} x_{li}(t)p_i + \bar{x}_{li}(t)q_i \end{aligned} \tag{2}$$

Where $w = 2/(n(n-1))$ is the probability that a particular pair of individuals is selected for an interaction and $\bar{w} = 2/n$ is the probability that a particular player is selected in an interaction. We use the shorthand notation $\bar{\varepsilon} := 1 - \varepsilon$ to denote the probability that no error was made while observing 3rd party interactions. We also use the shorthand notation $\bar{x}$ to denote $1 - x$. The above recursion covers all the cases in which i may update j's reputation after the interaction in round $t + 1$. Each of these cases corresponds to a line in Eq. (2).

1. In the first line, we cover the case where neither i nor j is selected for an interaction in time $t + 1$. This happens with probability $1 - 2\bar{w} + w$. In this case, i makes no change to j's reputation.

2. In the second line, we cover the case where both i and j were selected for an interaction in round t. This happens with probability w. In this interaction, j cooperates with probability $x_{ji}(t)$ and defects with with probability $1 - x_{ji}(t)$. If j cooperates, he is updated to *good* by i with probability p_i. If he defects, he is updated to *good* with probability q_i.

3. In the third line, we cover the case where j was part of an interaction but not i (this

happens with probability $\bar{w} - w$), *and* i chooses to ignore j's interaction (with probability $1 - \lambda_i$), therefore there is no update to j's reputation in round $t + 1$.

4. In the fourth line, we cover all the cases where j interacts with an arbitrary third party l (with probability w) and i takes this interaction into account (with probability λ_i). If j appears to cooperate ("appears" because i's observation is subject to errors), he is assigned *good* by i with probability p_i. Otherwise, if they appear to defect, they are assigned a *good* reputation with probability q_i.

5. In the fifth line, we cover the case where i but not j was part of an interaction (with probability $\bar{w} - w$) but i chose to not update j's reputation (with probability $1 - \gamma_i$).

6. In the sixth line, we cover all the cases where i has an interaction with third party l (with probability w) and she chooses to update j's reputation based on this interaction. If l cooperates with i, then j is assigned a *good* reputation with probability p_i and if l defects, j is assigned a *good* reputation with probability q_i.

By denoting $p_i - q_i := r_i$, one can simplify the recursion in Eq. (2) as,

$$\begin{aligned} x_{ij}(t+1) =& (1 - w - (\bar{w} - w)(\lambda_i + \gamma_i))x_{ij}(t) \\ &+ w r_i x_{ji}(t) \\ &+ w\lambda_i r_i (1 - 2\varepsilon) \sum_{l \neq i,j} x_{jl}(t) \\ &+ w\gamma_i r_i \sum_{l \neq i,j} x_{li}(t) \\ &+ w q_i + w\gamma_i q_i (n-2) + w\lambda_i (q_i + \varepsilon r_i)(n-2) \end{aligned} \tag{3}$$

2 Computation of expected payoffs

We collect all the probabilities x in a column vector of a size $n(n-1)$,

$$\mathbf{x}(t) := (x_{12}(t), x_{13}(t), \cdots, x_{1n}(t); \cdots; x_{n1}(t), x_{n2}(t), \cdots, x_{n(n-1)}(t))^{\intercal} \tag{4}$$

and define the square matrix $\mathbf{M} := (m_{ij,kl})$ and the column vector $\mathbf{v} := (v_{ij})$ as the following,

$$m_{ij,kl} = \begin{cases} 1 - w - (\bar{w} - w)(\lambda_i + \gamma_i) & \text{if } i = k, j = l \\ wr_i & \text{if } i = l, j = k \\ w\lambda_i r_i (1 - 2\varepsilon) & \text{if } j = k, \text{ and } i, j \neq l \\ w\gamma_i r_i & \text{if } i = l \text{ and } i, j \neq k \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

and,

$$v_{ij} = wq_i + w\gamma_i q_i (n-2) + w\lambda_i (q_i + \varepsilon r_i)(n-2). \tag{6}$$

Then, the recursion in Eq. (3) takes the following vector form,

$$\mathbf{x}(t+1) = \mathbf{M}\mathbf{x}(t) + \mathbf{v} \tag{7}$$

where $\mathbf{x}(0)$ is just the collection of the first elements, y, from players' strategies. That is, $x_{ik}(0) = y_i$ for all i and k. We can compute the average probability that players consider each other *good*, $\mathbf{x}$, by time discounting the vector $\mathbf{x}(t)$ in the following way,

$$\mathbf{x} := (1 - d) \sum_{t=0}^{\infty} d^t \mathbf{x}(t). \tag{8}$$

Now, by using the recursive relation in Eq. (7) and Eq (8), one can derive the following expression for the average probability vector, $\mathbf{x}$,

$$\mathbf{x} = (\mathbb{1} - d\mathbf{M})^{-1}((1 - d)\mathbf{x}_0 + d\mathbf{v}) \tag{9}$$

where $\mathbf{x}_0 = \mathbf{x}(0)$. One can interpret elements of $\mathbf{x}$ as the probability that players consider each other *good* (and therefore cooperate with others) in a randomly selected round. The expected payoff of player i is therefore,

$$\pi_i = \frac{1}{n-1} \sum_{j \neq i} x_{ji} b - x_{ij} c. \tag{10}$$

3 The Nash equilibria of the game

In this section we characterize all Nash equilibria strategies of the game with the form $(y, p, q, \lambda, \gamma)$. The strategy $\sigma := (y, p, q, \lambda, \gamma)$ is a Nash equilibrium strategy if no unilateral deviation by a single player in a homogeneous population playing σ leads the deviating player to a strictly better (expected) payoff. In our characterization of Nash strategies, we allow deviating players to deviate from σ to *any* arbitrary strategy, even strategies that cannot be represented by a 5-tuple. We first present some auxiliary results.

Proposition 1. *(Cooperation and payoffs in homogenous population) Consider a population of size n where every individual plays $(y, p, q, \lambda, \gamma)$ in a game where $\delta < 1$. Let $r := p - q$. Then, the average probability that players cooperate with each other is*

$$x = \frac{(1-\delta)y + \delta q + \delta(n-2)(q(\lambda+\gamma) + \lambda\varepsilon r)}{1 - \delta r + \delta(n-2)((\lambda+\gamma)(1-r) + 2\lambda\varepsilon r)}. \tag{11}$$

In this homogenous population each player receives an average payoff, $\pi = (b-c)x$.

Further, if the game is generic, i.e, $n > 2, \varepsilon > 0$, a population is

1. *fully defective ($x = 0$) if and only if, $y = 0$, $q = 0$ and, in addition, $\lambda = 0$ or $p = 0$.*
2. *fully cooperative ($x = 1$) if and only if, $y = 1, p = 1$ and, in addition, $\lambda = 0$ or $q = 1$.*

If the game is non-generic, i.e., $n = 2$ or $\varepsilon = 0$, a population is,

1. *fully defective if and only if $y = q = 0$.*
2. *fully cooperative if and only if $y = p = 1$.*

All proofs of propositions, lemmas and theorems are in the appendix.

Now consider a population where players $1, 2, ..., n-1$ are playing the same strategy $\sigma = (y, p, q, \lambda, \gamma)$ while player n is playing an arbitrary strategy. For simplicity, we call these $n-1$ players as *residents* and the last player, n as *mutant*. In the following lemmas we derive some linear relationships between the average rates of cooperation in this population,

Lemma 1. *Consider $n > 2$, and the game with pairwise continuation probability, δ, and observation error probability, ε. Let $i, k \in \{1, 2, ..., n-1\}$ such that $i \neq k$. All players in $\{1, 2, ..., n-1\}$ play the strategy $(y, p, q, \lambda, \gamma)$. Then, the following linear relationships are satisfied, independent of the arbitrary strategy that player n adopts,*

$$\begin{aligned} x_{in} &= B_1 + B_2 x_{ni} + B_3 x_{ik} \\ x_{ik} &= C_1 + C_2 x_{ni} + C_3 x_{in} \end{aligned} . \tag{12}$$

The coefficients B_1, B_2, B_3 and C_1, C_2, C_3 are given by

$$\begin{aligned} B_1 &= \frac{(1-\delta)y + \delta q + \delta(n-2)(q(\lambda+\gamma) + \lambda\varepsilon r)}{1 + \delta(n-2)(\lambda+\gamma)} \\ B_2 &= \frac{\delta r(1 + \lambda(n-2)(1-2\varepsilon))}{1 + \delta(n-2)(\lambda+\gamma)} \\ B_3 &= \frac{\delta\gamma r(n-2)}{1 + \delta(n-2)(\lambda+\gamma)} \\ C_1 &= \frac{(1-\delta)y + \delta q + \delta(n-2)(q(\lambda+\gamma) + \lambda\varepsilon r)}{1 - \delta r + \delta(\lambda+\gamma)(n-2) - \delta r(n-3)(\lambda+\gamma-2\lambda\varepsilon)} \\ C_2 &= \frac{\delta\gamma r}{1 - \delta r + \delta(\lambda+\gamma)(n-2) - \delta r(n-3)(\lambda+\gamma-2\lambda\varepsilon))} \\ C_3 &= \frac{\delta\lambda r(1-2\varepsilon)}{1 - \delta r + \delta(\lambda+\gamma)(n-2) - \delta r(n-3)(\lambda+\gamma-2\lambda\varepsilon)} \end{aligned} \tag{13}$$

The average probability that a resident cooperates with player n, x_{in}, varies linearly with the average probability that residents cooperate with residents x_{ik}, and player n cooperates with residents, x_{ni}. Using Lemma 1 we can show that a linear relationship exists between x_{in} and x_{ni} for any strategy that the mutant adopts,

Proposition 2. *Consider the game with n players, pairwise continuation probability, δ, and observation error probability ε. All players in set $\{1, 2, ..., n-1\}$ play the strategy $(y, p, q, \lambda, \gamma)$ and player n plays an arbitrary strategy. Then, the following relationship between average cooperation shown by residents to the mutant, x_{in}, and the average cooperation shown by the mutant to the residents, x_{ni}, is satisfied irrespective of the strategy the mutant (player n) adopts,*

$$x_{in} = K_1 + K_2 x_{ni}. \tag{14}$$

The coefficients K_1 and K_2 can be expressed in terms of the coefficients given in Eq. (13) as,

$$K_1 = \begin{cases} \dfrac{B_1 + B_3 C_1}{1 - B_3 C_3} & \text{if } n > 2 \\ (1-\delta)y + \delta q & \text{if } n = 2 \end{cases}. \tag{15}$$

$$K_2 = \begin{cases} \dfrac{B_2 + B_3 C_2}{1 - B_3 C_3} & \text{if } n > 2 \\ \delta r & \text{if } n = 2 \end{cases}. \tag{16}$$

As a result, the mutant's payoff can be written as,

$$\pi_n = K_1 b + (K_2 b - c) x_{ni} \tag{17}$$

The proposition above demonstrates that the resident strategy enforces a linear relationship on the payoffs of the mutant to how much the mutant cooperates with them (this is similar to zero-determinant strategies in the literature of direct reciprocity[2]). When the slope, $K_2 b - c < 0$ is set to be strictly negative by the resident strategy, the best response for the mutant is to always defect. Similarly, when the slope is set to be strictly positive, the best response is to always cooperate. When the slope is exactly zero, any strategy that the mutant adopts is a best response as all strategies earn the same payoff of $K_1 b$. In this case, the resident strategy is also a best-response to itself. While characterizing all Nash equilibria strategies of the form $(y, p, q, \lambda, \gamma)$, we study the sign of the slope, $K_2 b - c$, of this linear relationship. In the next lemma, we compare the signs of the slope $K_2 b - c$ and a quadratic in $r := p - q$. This simplifies the characterization of Nash strategies in the next section.

Lemma 2. *Consider $n > 2$. Then, if we define the coefficients $\Lambda_1, \Lambda_2, \Lambda_3$ of the quadratic in r, $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3$ in the following way,*

$$\begin{aligned}
\Lambda_1 &= \delta^2\gamma(n-2)(\gamma b + \lambda c(1-2\varepsilon)) - b\delta D(1+\lambda(n-2)(1-2\varepsilon)) \\
\Lambda_2 &= b\delta A(1+\lambda(n-2)(1-2\varepsilon)) + cAD \\
\Lambda_3 &= cA^2
\end{aligned} \tag{18}$$

where A and D are given by,

$$\begin{aligned}
A &= 1 + \delta(n-2)(\lambda+\gamma) \\
D &= \delta(1 + (n-3)(\lambda+\gamma-2\lambda\varepsilon))
\end{aligned} \tag{19}$$

then, the following statements are true,

1. $K_2 b - c = 0 \Leftrightarrow \Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 = 0$
2. $K_2 b - c > 0 \Leftrightarrow \Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 > 0$
3. $K_2 b - c < 0 \Leftrightarrow \Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 < 0$

where K_2 is given by Eq. (16).

The above lemma shows that the signs of $K_2 b - c$ and the quadratic in r, $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3$ are the same when $n > 2$. The result from this lemma allows us to focus only on the properties of the quadratic in r instead of studying the properties of $K_2 b - c$ while characterizing the Nash equilibria.

Characterization of Nash equilibria

In the two following theorems, we characterize all Nash equilibria of the form $(y, p, q, \lambda, \gamma)$ for games where $\delta < 1$.

Theorem 1. *(characterization of Nash equilibria for $n > 2$)*
For $n > 2$ and $\delta < 1$, a strategy $(y, p, q, \lambda, \gamma)$ is a Nash equilibrium if and only if atleast one of the following conditions holds,

1. *The strategy always defects. That is, $y = p = q = 0$ (a generic Nash)*

2. *It is an equalizer strategy that sets a mutant's payoff to a constant value. That is,* $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 = 0$ *(a generic Nash)*

3. $y = p = 1$ ***and*** $\Lambda_1 (1-q)^2 + \Lambda_2 (1-q) - \Lambda_3 > 0$ ***and*** $\varepsilon = 0$ *or* $\lambda = 0$

4. $y = q = 0$ ***and*** $\Lambda_1 p^2 + \Lambda_2 p - \Lambda_3 < 0$ ***and*** $\varepsilon = 0$ *or* $\lambda = 0$.

where $r := p - q$ *and* Λ_1, Λ_2 *and* Λ_3 *are given by Eq. (18) and Eq. (19).*

We call the set containing ALLD and all equalizer strategies (the strategies that set mutant's payoff to a constant value), the set of *generic* Nash equilibria.

Theorem 2. *(characterization of Nash equilibria for* $n = 2$*)*
For $n = 2$ *and* $\delta < 1$*, a strategy* $(y, p, q, \lambda, \gamma)$ *is a Nash equilibrium of the game if and only if one of the following is true:*

1. $p - q = c/(b\delta)$,

2. $y = q = 0$ *and* $p < c/(b\delta)$

3. $y = p = 1$ *and* $q < 1 - c/(b\delta)$

The theorem above states the Nash equilibria for games with only two players in the population. In this case, by definition, indirect and generalized reciprocity are absent.

Cooperative Nash equilibria

The theorems in the previous subsection fully characterize the set of all Nash equilibria. In this subsection, we study the subset of all Nash equilibria which sustain full cooperation in the population in the limit where errors are absent.

Definition 1. *(generic cooperative Nash equilibrium): A Nash equilibrium strategy,* σ*, is a (generic) cooperative Nash equilibrium when it is*

1. *a generic Nash equilibrium and,*

2. *in a homogenous population playing* σ*, the cooperation rate according to Eq. (11) approaches one in the limit of rare errors. That is* $x \to 1$ *as* $\varepsilon \to 0$.

The definition above distinguishes between a cooperative Nash equilibrium and a *generic* cooperative Nash equilibrium. Generic cooperative Nash equilibria are a subset of cooperative Nash equilibrium. They require that the Nash equilibrium strategies are fully cooperative between themselves and also equalize the payoff of any mutant.

Proposition 3. *(characterization of generic cooperative Nash equilibria)*
For a game where $\delta < 1$, a strategy $(y, p, q, \lambda, \gamma)$ is a generic cooperative Nash equilibrium if and only if

1. $y = p = 1$, *and,*
2. $\Lambda_1(1-q)^2 + \Lambda_2(1-q) - \Lambda_3 = 0$

The conditions for a strategy with only direct and indirect reciprocity to be a cooperative Nash equilibria have been studied in detail in the original model from Schmid et al.[1]. In the following propositions, we present properties of generic cooperative Nash equilibrium with direct (DR) and generalized reciprocity (GR) only ($\lambda = 0$). In these propositions, we naturally assume $n > 2$ since the conditions involve generalized reciprocity.

Proposition 4. *(Existence and uniqueness of a generic cooperative Nash with pure GR)*
A strategy of the form $(1, 1, q^*_{\mathrm{G}}, 0, 1)$ is a cooperative Nash equilibrium if and only if $\delta > c/b$ and,

$$q^*_{\mathrm{G}} = 1 - \frac{c + c\delta(n-2)}{b\delta + c\delta(n-2)}. \tag{20}$$

Interestingly, the condition $\delta > c/b$ is the same necessary condition as in the case of direct reciprocity. However, for a fixed value of $\delta > c/b$, the generosity, q^*_{D}, at a pure DR equilibrium is strictly greater than the generosity, q^*_{G}, at a pure GR equilibrium,

$$1 - \frac{c}{b\delta} = q^*_{\mathrm{D}} > q^*_{\mathrm{G}}. \tag{21}$$

Proposition 5. *(Existence of a unique generic cooperative Nash with only DR and GR)*
Consider a $\gamma \in [0, 1]$. Then, if and only if the following implicit condition is satisfied,

$$\delta(1-\gamma)((n-2)\gamma + 1) \leq (1 + \delta(n-2)\gamma)\left(1 + \frac{c}{b} - \frac{c\gamma}{b} - \frac{c}{b\delta}\right) \tag{22}$$

then there is a unique $q^* \in [0,1)$ *such that the strategy* $(1,1,q^*,0,\gamma)$ *is a generic cooperative Nash equilibrium.*

The above proposition lays down a necessary and sufficient condition on the pairwise continuation probability, δ, for the existence of a unique generic cooperative Nash equilibria with a specific mixture of direct and generalized reciprocity (determined by the strategy element γ) and no indirect reciprocity (strategy element $\lambda = 0$). By plugging in $\gamma = 0$ and $\gamma = 1$ in the implicit condition above, one recovers identical necessary conditions, $\delta > c/b$.

Proposition 6. *Consider a* $\gamma \in [0,1]$. *If* $(1,1,q^*,0,\gamma)$ *is the unique generic cooperative Nash equilibrium of the repeated game, then for any* $q \in [0,q^*)$, $(1,1,q,0,\gamma)$ *is a (non-generic) cooperative Nash equilibrium. Furthermore, there is no* $q > q^*$ *such that* $(1,1,q,0,\gamma)$ *is a cooperative Nash equilibrium.*

This proposition is about the set of cooperative Nash equilibrium strategies that do not use indirect reciprocity ($\lambda = 0$) and only use a particular mixture of direct and generalized reciprocity (determined by their shared γ value). It claims that if this set contains a unique generic Nash equilibrium, then, the only other element in this set is a continuum of non-generic Nash equilibrium. In addition, all Nash equilibrium in this continuum display a generosity q strictly less than the generosity of the generic Nash equilibrium q^*.

Q: What is the minimum pairwise continuation probability necessary for a combination of direct and generalized reciprocity to support a cooperative equilibrium?

A: The minimum continuation probability needed for direct or generalized reciprocity to individually support a cooperative equilibrium (c/b) is too low for a combination of them to do the same. We find that the minimum continuation probability necessary for a DR-GR mixture to form a Nash equilibrium depends on how much weight is put on each reciprocity (i.e., the value of γ). More precisely, with numerical calculations, we find that the condition in Eq. (22) must be satisfied by the pairwise continuation probability δ.

It was observed in Schmid et al. that if both direct and indirect reciprocity are able to independently constitute a cooperative Nash equilibrium, any possible mixture of them can do the same[1]. Interestingly, an equivalent statement involving direct and generalized reciprocity does not hold. We show that for certain $\delta > c/b$, it is possible that direct and generalized reciprocity

are able to independently stabilize cooperation but a strategy that uses a mixture of them cannot. To elucidate this result, we perform a deviation analysis (Supplementary Fig. S1). We imagine a single individual deviating to pure defection (the mutant) from an otherwise homogenous resident population. In the resident population, everyone plays the strategy $(1, 1, 0.01, 0, \gamma)$. When residents only use direct reciprocity ($\gamma = 0$), they cooperate with other residents but rarely forgive the defector. As a result the mutant's payoff is below $b - c$, the payoff of the homogenous resident population. In contrast, when residents only use generalized reciprocity ($\gamma = 1$), cooperative residents treat everyone equally but they almost always defect. Also here the mutant receives less than $b - c$. However, when residents use a mixture of the two kinds of reciprocity ($0 < \gamma < 1$), switching to defection can be strictly beneficial (Supplementary Fig. S1B).
As a result, stable cooperation via a mixture of direct and generalized reciprocity requires more interactions than stable cooperation by either pure direct or pure generalized reciprocity (Supplementary Fig S1C).

4 Effective likelihood of direct, indirect and generalized reciprocity

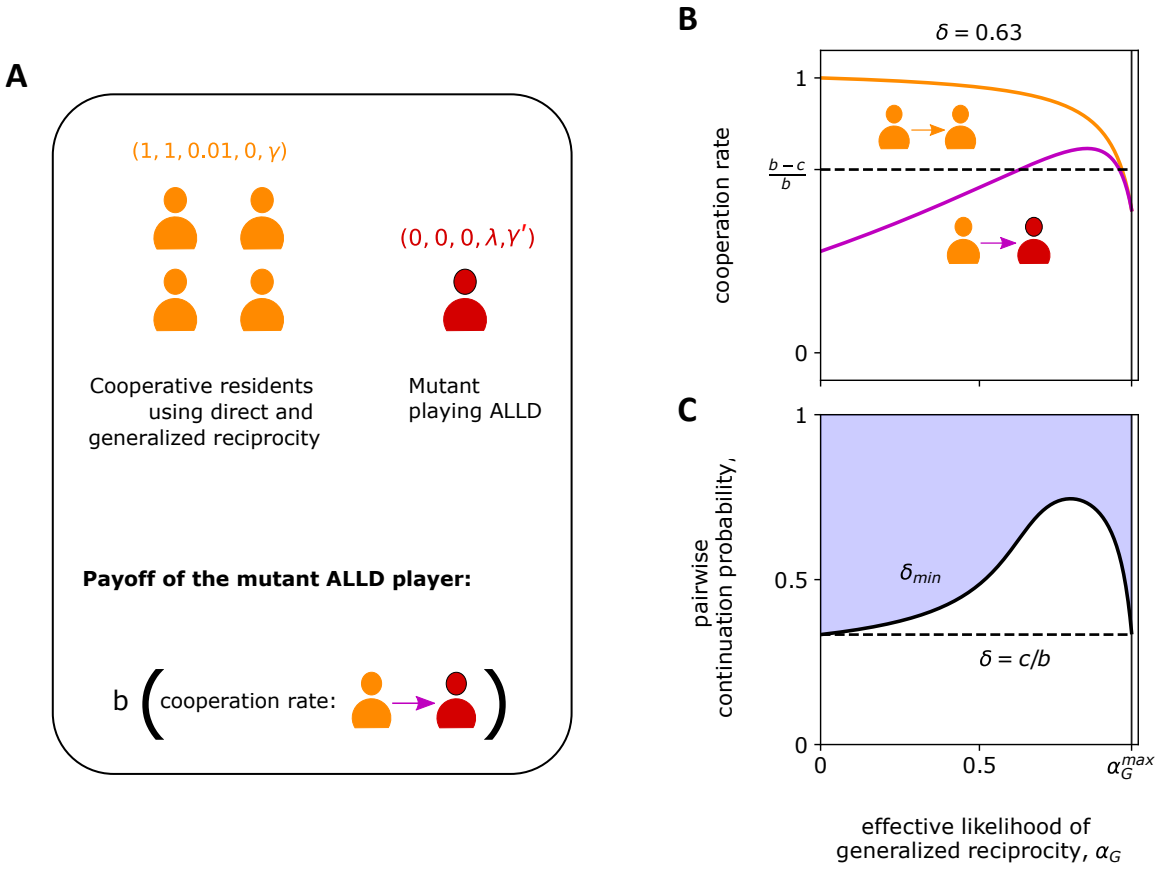
In this section we introduce three probabilities to describe the effective likelihood with which players use direct, indirect and generalized reciprocity in a homogenous population. We define these probabilities below,

1. **The effective likelihood of direct reciprocity**, α_{D}: the probability that the reputation of an arbitrary individual j from the perspective of another arbitrary individual i is not altered between two direct interactions involving i and j.

2. **The effective likelihood of indirect reciprocity**, α_{I}: the probability that between two direct interactions involving i and j, i updates j's reputation *and*, the last update he makes about j's reputation before meeting her is due to j's interaction with a third party $k \neq i$.

3. **The effective likelihood of generalized reciprocity**, α_{G}: the probability that between two direct interactions involving i and j, i updates j's reputation *and*, the last update he makes about j's reputation before meeting her is due to i's interaction with a third party $k \neq i$.

By definition then $\alpha_{\mathrm{D}} + \alpha_{\mathrm{I}} + \alpha_{\mathrm{G}} = 1$. This normalizing condition allows us to quantify how much weight players in a homogenous population put on direct, indirect and generalized reciprocity. In a population where all individuals play $(y, p, q, \lambda, \gamma)$, the effective likelihood of direct reciprocity, α_{D}, can be calculated as



Supplementary Figure S1: A mixture of direct and generalized reciprocity may be less conducive to cooperation than either pure direct or pure generalized reciprocity (A), We consider the situation where an always defecting (ALLD) mutant is introduced into a homogenous population of cooperative residents who play the strategy $(1, 1, q, 0, \gamma)$. Before the mutant is introduced, the residents receive an expected payoff of $b - c$. To be a Nash equilibrium, the average probability that residents cooperate with the mutant must not exceed $(b - c)/b$. **(B),** The way the residents react to each other (orange curve) and to the ALLD mutant (pink curve) depends on how often they use generalized reciprocity to update reputations, α_{G}. When the residents use pure direct reciprocity, i.e., $\alpha_{\mathrm{G}} = 0$, they employ generous tit-for-tat and discriminate against the ALLD mutant. They cooperate fully with other residents but rarely forgive the mutant. As α_{G} increases, the two curves tend to converge. When residents use pure generalized reciprocity, i.e., $\alpha_{\mathrm{G}} = \alpha_{\mathrm{G}}^{max}$, they treat everyone equally. If the pairwise continuation probability, δ, is not sufficiently high (for example in this case where $\delta = 0.63$), the ALLD mutant earns a payoff greater than $b - c$ for certain $0 < \alpha_{\mathrm{G}} < 1$. **(C),** Depending on α_{G}, the minimum value of the pairwise continuation probability, δ_{min}, that is necessary for a resident strategy to be a best response to itself, varies. It is at its lowest, $\delta_{min} = c/b$, when residents use pure direct or pure generalized reciprocity. Parameters: $b = 3$, $c = 1$ and $n = 50$.

$$\alpha_{\mathrm{D}} = \frac{1}{\binom{n}{2}} \cdot \sum_{s=0}^{\infty} \sum_{l+m=0}^{s} \frac{s\,!}{l\,!m\,!(s-l-m)\,!} \cdot \zeta_1^{l+m} \cdot \zeta_2^{s-l-m} \cdot (1-\lambda)^l (1-\gamma)^m$$

$$= \frac{1}{1+(n-2)(\lambda+\gamma)}. \tag{23}$$

Where ζ_1 is the probability that either i or j (not both) is part of an interaction at a particular round and ζ_2 is probability that neither i nor j is part of a particular interaction. These probabilities are given by,

$$\zeta_1 = \frac{n-2}{\binom{n}{2}},$$

$$\zeta_2 = \frac{\binom{n-2}{2}}{\binom{n}{2}}. \tag{24}$$

The probability that between two direct interactions involving i and j, the reputation of j with respect to i is not updated depends on the likelihood of two events. Either neither of them are involved in an interaction between these two interactions (in this way there is no opportunity to update reputations) or, one of them is part of an interaction, but the reputation consequences for j because of this interaction are ignored. In the expression inside the sums in Eq. (23), we consider that out of the s interactions between two direct interactions involving i and j, l interactions involve only j, m interactions involve only i, and, $s-l-m$ interactions involve neither i nor j. These events jointly occur with probability from the multinomial distribution,

$$\frac{s\,!}{l\,!m\,!(s-l-m)\,!} \cdot \zeta_1^{l+m} \zeta_2^{s-l-m}$$

.

In all the l interactions in which only j is involved, i ignores j's action with probability $1-\lambda$. Similarly in all the m interactions involving only i, i does not update j's reputation based on what is done to him, with probability $1-\gamma$. Thus, the probability that i makes no update to j's reputation in this exact distribution of events (i.e., for particular values of s, l and m) is,

$$\frac{s\,!}{l\,!m\,!(s-l-m)\,!} \cdot \zeta_1^{l+m} \cdot \zeta_2^{s-l-m} \cdot (1-\lambda)^l (1-\gamma)^m.$$

To compute α_{D} we sum up the above probability for all values of l and m that sum up to s and then sum for all values of s. The probability $1/(\binom{n}{2})$ on the outside of the sum in Eq. (23) is the probability that i and j are chosen for their second direct interaction.

The probability $1-\alpha_{\mathrm{D}}$ is the probability that the reputation of j from the perspective of i is updated between two direct interactions involving i and j. The final reputation of j from i's perspective before they meet could be because of an interaction that involved j but not i (indirect reciprocity) or an interaction that involved i but not j (generalized reciprocity). Then, the likelihood α_{I} and α_{G} can be expressed as

$$\begin{aligned} \alpha_{\mathrm{I}} &= (1-\alpha_{\mathrm{D}}) \cdot \zeta_3 \cdot \lambda, \\ \alpha_{\mathrm{G}} &= (1-\alpha_{\mathrm{D}}) \cdot \zeta_3 \cdot \gamma. \end{aligned} \tag{25}$$

where ζ_3 is a constant of proportionality. Now as, $\alpha_{\mathrm{I}} + \alpha_{\mathrm{G}} = 1 - \alpha_{\mathrm{D}}$,

$$(1-\alpha_{\mathrm{D}}) = (1-\alpha_{\mathrm{D}}) \cdot \zeta_3 \cdot (\lambda + \gamma) \tag{26}$$

Since $1-\alpha_{\mathrm{D}}$ is non-zero only when $\lambda, \gamma > 0$,

$$\alpha_{\mathrm{I}} = \begin{cases} (1-\alpha_{\mathrm{D}}) \cdot \frac{\lambda}{\lambda+\gamma} & \text{if } \lambda+\gamma \neq 0 \\ 0 & \text{otherwise} \end{cases} \tag{27}$$

and,

$$\alpha_{\mathrm{G}} = \begin{cases} (1-\alpha_{\mathrm{D}}) \cdot \frac{\gamma}{\lambda+\gamma} & \text{if } \lambda+\gamma \neq 0 \\ 0 & \text{otherwise.} \end{cases} \tag{28}$$

Theoretical minimum of the likelihood of direct reciprocity: In this model, as long as $\delta > 0$, there is a non-zero probability that direct reciprocity is employed in updating reputations. This is because there is always a non-zero chance that a pair of players will meet again after they meet in the present *and* in-between their two direct interactions, one of them will not update the other's reputation (either because neither of them participate in an interaction or the player choose to ignore the other's interactions). From Eq. (23), we can see that this theoretical

minimum occurs when $\lambda + \gamma = 1$,

$$\alpha_{\mathrm{D}}^{\min} = \frac{1}{n-1}. \tag{29}$$

This also imposes a theoretical maximum on the likelihood of indirect reciprocity and the likelihood of generalized reciprocity which occur at $(\lambda, \gamma) = (1, 0)$ and $(0, 1)$ respectively,

$$\alpha_{\mathrm{I}}^{\max} = \alpha_{\mathrm{G}}^{\max} = 1 - \alpha_{\mathrm{D}}^{\min} = \frac{n-2}{n-1}. \tag{30}$$

Any tuple of the effective likelihoods, $(\alpha_{\mathrm{D}}, \alpha_{\mathrm{I}}, \alpha_{\mathrm{G}})$ can be expressed as a convex combination of the corners of the triangular simplex, $\mathcal{S}_{\mathrm{DIG}}(n)$, formed by the points: $(1, 0, 0)$, $(\alpha_{\mathrm{D}}^{\min}, \alpha_{\mathrm{I}}^{\max}, 0)$, $(\alpha_{\mathrm{D}}^{\min}, 0, \alpha_{\mathrm{G}}^{\max})$. For a fixed population size, n, every tuple $(\lambda, \gamma) \in [0, 1]^2$ maps to a unique point inside the simplex, $\mathcal{S}_{\mathrm{DIG}}(n)$. Formally, every triplet $(n, \lambda, \gamma) \in \mathbb{Z}_{\geq 2} \times [0, 1]^2$ has a unique map in $\mathcal{S}_{\mathrm{DIG}}(n)$. That is, there is a one-to-one mapping between the two spaces.

5 Numerically efficient computation of payoffs

For evolutionary simulations, we numerically compute the players' payoffs. The payoffs in a homogenous population can be explicitly computed using the result in Proposition 1. However, in heterogenous populations, one needs to calculate the payoffs of players using Eq. (10). This requires computing the inverse of a matrix of size $n(n-1)$, which could result in large computation times for large populations. In this section we discuss a more efficient way to compute payoffs. This efficient way takes advantage of the fact that players using the same strategy have the same reputation from the perspective of all other players.

We consider that there are s distinct strategies in the population (indexed from 1 to s) and the sizes of these sub-populations are $k_1, k_2, .., k_s$. The i-th strategy plays $\sigma_i := (y_i, p_i, q_i, \lambda_i, \gamma_i)$. The recursion equation describing the change in the probability that players consider each other *good* Eq. (3), can now be written to describe the change in the probability that *strategies* consider each other *good*. In the following recursions, $x_{ij}(t)$ indicates the probability that at time t, a player using strategy i considers a player using strategy j as *good*,

$$
\begin{aligned}
x_{ii}(t+1) =& (1-w-(\bar{w}-w)(\lambda_i+\gamma_i)+wr_i+(k_i-2)wr_i\lambda_i(1-2\varepsilon)+(k_i-2)wr_i\gamma_i)x_{ii}(t) \\
&+ w\lambda_i r_i(1-2\varepsilon)\sum_{l\neq i}^{s} k_l x_{il}(t) \\
&+ w\gamma_i r_i \sum_{l\neq i}^{s} k_l x_{li}(t) \\
&+ wq_i + w\gamma_i q_i(n-2) + w\lambda_i(q_i+\varepsilon r_i)(n-2) \\
\\
x_{ij}(t+1) =& (1-w-(\bar{w}-w)(\lambda_i+\gamma_i))x_{ij}(t) \\
&+ (wr_i + w\lambda_i r_i(1-2\varepsilon)(k_i-1) + w\gamma_i r_i(k_j-1))x_{ji} \\
&+ w\lambda_i r_i(1-2\varepsilon)\sum_{l\neq i}^{s}(k_l - 1_{jl})x_{jl}(t) \\
&+ w\gamma_i r_i \sum_{l\neq i,j}^{s} k_l x_{li}(t) \\
&+ w\gamma_i r_i(k_i-1)x_{ii} \\
&+ wq_i + w\gamma_i q_i(n-2) + w\lambda_i(q_i+\varepsilon r_i)(n-2)
\end{aligned}
\tag{31}
$$

Here $r_i := p_i - q_i$ and 1_{ij} is the Kronecker delta function which is 1 if $i = j$ and 0 if $i \neq j$. In our evolutionary simulations we always compute payoffs for populations with at most two strategies (that is, in the limit of rare mutations in the Imhof-Nowak process[3,4]). We describe the remaining steps of the efficient computation method for only this special case. However, one can already see that recursions have a general form for any s in Eq. (31). Using the same technique, it is also possible to construct the remaining steps for arbitrary number of strategies in the population.

We assume that the two strategies in the population are $\sigma_1 = (y_1, p_1, q_1, \lambda_1, \gamma_1)$ and $\sigma_2 = (y_2, p_2, q_2, \lambda_2, \gamma_2)$. The expected probability that strategies consider each other *good* at a randomly selected round can be computed as,

$$
\mathbf{x} = (\mathbb{1} - d\mathbf{M})^{-1}((1-d)\mathbf{x_0} + d\mathbf{v}) \tag{32}
$$

Here $\mathbf{x}$ is a column vector of size 4,

$$\mathbf{x} = \begin{pmatrix} x_{11} \\ x_{12} \\ x_{21} \\ x_{22} \end{pmatrix} \tag{33}$$

and the 4×4 square matrix $\mathbf{M} = (m_{ab,cd})_{a,b,c,d \in \{1,2\}}$, the four-dimensional vectors $\mathbf{x}_0$ and $\mathbf{v}$ are given by

$$m_{ab,cd} = \begin{cases} 1 - w + wr_a - (\bar{w} - w)(\lambda_a + \gamma_a) & \\ +wr_a(k_a - 2)\lambda_a(1 - 2\varepsilon) + wr_a(k_a - 2)\gamma_a & \text{if } a = b = c = d, \\ 1 - w - (\bar{w} - w)(\lambda_a + \gamma_a) & \text{if } a = c \neq b = d, \\ 0 & \text{if } a = b \neq c = d, \\ w\lambda_a r_a(1 - 2\varepsilon)k_d & \text{if } a = b = c \neq d, \\ w\gamma_a r_a k_c & \text{if } a = b = d \neq c, \\ w\gamma_a r_a(k_a - 1) & \text{if } a = c = d \neq b, \\ wr_a + wr_a\gamma_a(k_b - 1) & \\ +wr_a\lambda_a(1 - 2\varepsilon)(k_a - 1) & \text{if } a = d \neq b = c, \\ w\lambda_a r_a(1 - 2\varepsilon)(k_b - 1) & \text{if } a \neq b = c = d. \end{cases} \tag{34}$$

$$\mathbf{x_0} = \begin{pmatrix} y_1 \\ y_1 \\ y_2 \\ y_2 \end{pmatrix}, \tag{35}$$

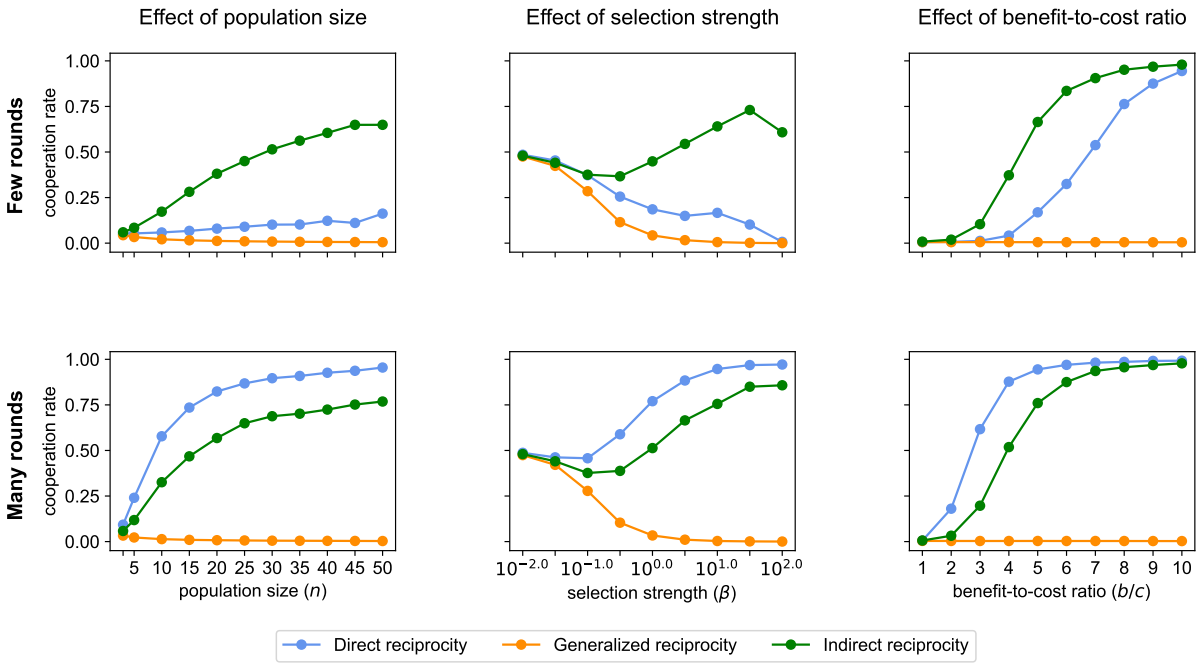
$$\mathbf{v} = \begin{pmatrix} wq_1 + w\gamma_1 q_1(n-2) + w\lambda_1(q_1 + \varepsilon r_1)(n-2) \\ \\ wq_1 + w\gamma_1 q_1(n-2) + w\lambda_1(q_1 + \varepsilon r_1)(n-2) \\ \\ wq_2 + w\gamma_2 q_2(n-2) + w\lambda_2(q_2 + \varepsilon r_2)(n-2) \\ \\ wq_2 + w\gamma_2 q_2(n-2) + w\lambda_2(q_2 + \varepsilon r_2)(n-2) \\ \\ \end{pmatrix} \tag{36}$$

respectively. Payoff for players using strategy i can then be calculated from the computed $\mathbf{x}$ from Eq. (32) using the expression,

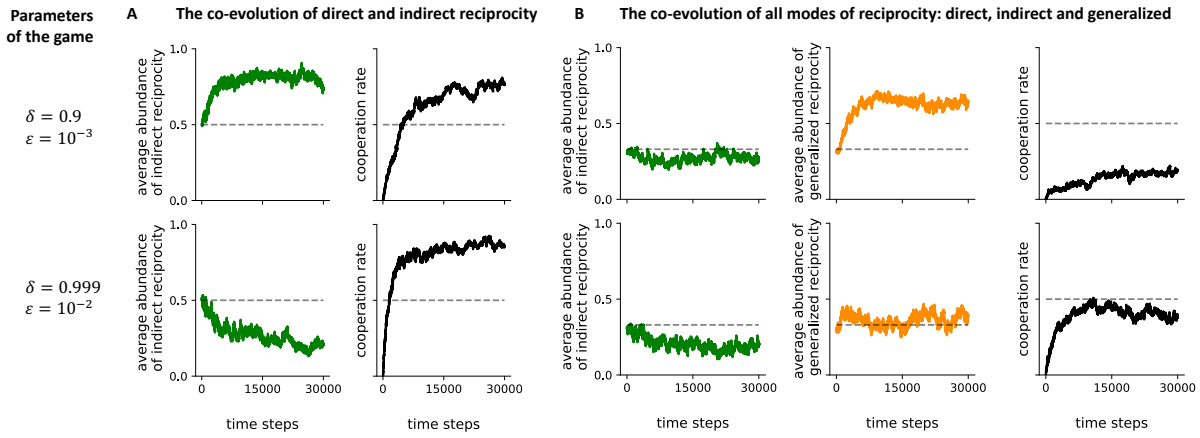
$$\pi_i = \sum_{j=1}^{s} \frac{k_j - 1_{ij}}{n-1}(x_{ji}b - x_{ij}c). \tag{37}$$

For s strategies in the population, the method of efficiently computing payoffs requires the computation of the inverse of a matrix of size s^2 (as opposed to the original method where the matrix size was $n(n-1)$).

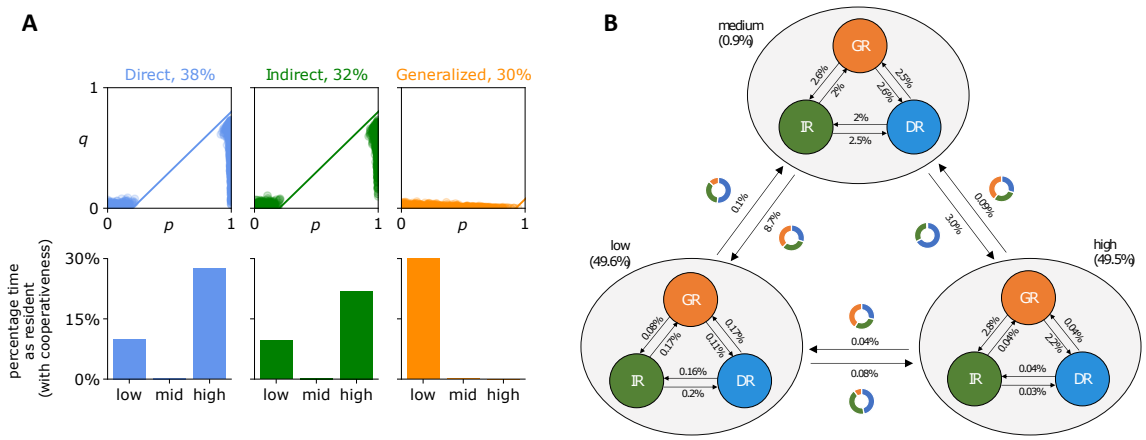
Supplementary Figures



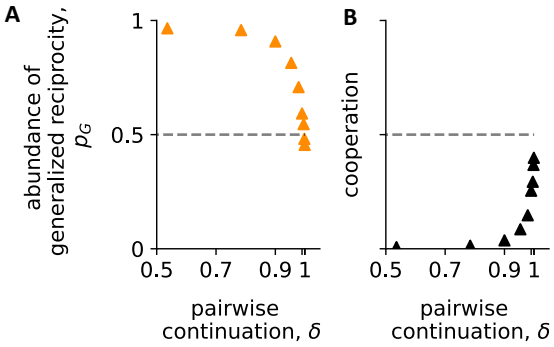
Supplementary Figure S2: The evolution of cooperation when populations rely on a single mode of reciprocity and rarely explore. We explore how the effect of Figure 3 (from main text) is influenced by changing different parameters of the model. Specifically, we examine the effects of population size, selection strength, and the benefit-to-cost ratio. We consider two scenarios: one with a few rounds ($\delta = 0.5$, top row) and one with many rounds ($\delta = 1$, bottom row). Each data point represents the average cooperation rate of the simulation process. In the bottom panel, with many rounds, it is the average of a single simulation. For the top panel, it is the average of three independent simulations. Each simulation stops after 10^7 mutant strategies have been introduced into the population. If a parameter is not varying, it is fixed to the same value we use for Figure 3. That is, population size $n = 50$ and benefit of cooperation, $b = 5$, cost of cooperation, $c = 1$, and selection strength, $\beta = 10$. For all these simulations, we assume that there is no error in observing third party interactions, i.e, $\varepsilon = 0$. We observe that cooperation fails to evolve via generalized reciprocity across a range of values of n, β and b/c.



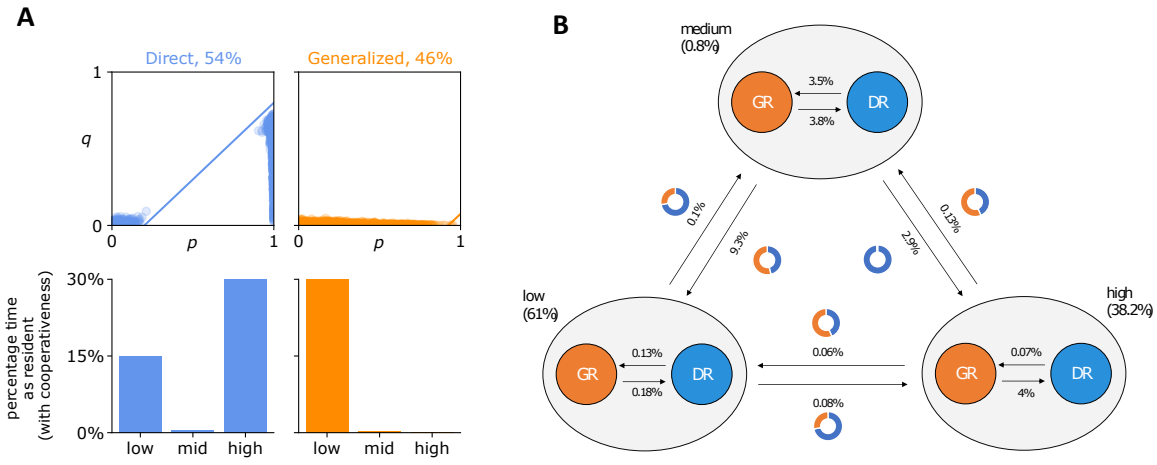
Supplementary Figure S3: Evolutionary dynamics with and without generalized reciprocity demonstrates the impact of generalized reciprocity on cooperation. Here we present average evolutionary trajectories for two kinds of evolutionary dynamics as in Fig.4 from main text. We show the time series of the likelihood that players use indirect reciprocity (α_{I}), the likelihood that they use generalized reciprocity (α_{G}) in a population, averaged over 150 independent simulations. In simulations performed for **(A)**, individuals can only adopt direct and indirect reciprocity. However in **(B)**, they can adopt all the three modes of reciprocity. We also plot the average cooperation in populations as they evolve over time. We show these time series for two parameter combinations: $\delta = 0.9, \varepsilon = 10^{-3}$ (top row) and $\delta = 0.999, \varepsilon = 10^{-2}$ (bottom row). Cooperation rates remain considerably low in the evolutionary process where individuals can adopt generalized reciprocity.



Supplementary Figure S4: The analysis of a simulation with many pairwise interactions and rare observational errors reveals a mechanism behind the breakdown of cooperation in the presence of generalized reciprocity. We conduct an evolutionary simulation at low mutation rate with the parameters $\delta = 0.999$, and $\varepsilon = 10^{-3}$. Individuals in the simulation were free to adopt any of the three modes of reciprocity. **(A),** Among the 47,000 resident populations that emerged, 38% used direct reciprocity, 32% used indirect reciprocity and the rest, 30%, used generalized reciprocity. We show the strategy elements p and q of the 1000 most successful residents from each reciprocity class using colored dots. We categorize each resident population into one of the three classes: low (less than 1/3), medium (between 1/3 and 2/3) and high (more than 2/3), based on the probability of observing cooperation in them. The most abundant residents are high DR and low GR. **(B),** In this illustration we show the estimated transition probabilities between the cooperative classes (arrows between the 3 large ovals) and estimated transition probabilities between residents that use different modes of reciprocity but fall under the same cooperative class (arrows between the colored circles inside each oval). We estimate these transition probabilities using data from our simulation. For each transition between the cooperative classes, we denote the relative contribution of each mode of reciprocity for that transition using colors on a circular scale (blue for direct reciprocity, green for indirect reciprocity and orange for generalized reciprocity). All other parameters are the same as in Fig. 3 of main text.



Supplementary Figure S5: The evolutionary dynamics of direct and generalized reciprocity. In the evolutionary simulations that we conduct for this Figure, individuals can only choose to adopt either pure direct reciprocity or pure generalized reciprocity. The evolutionary process is conducted for rare mutations. In all simulations, the evolutionary process begins with a strategy that always defects. In **(A)**, we show the average time series of the likelihood that individuals use indirect reciprocity, α_{I} (which is trivially always zero since we restrict our simulations to exclude indirect reciprocity) and the likelihood that they use generalized reciprocity, α_{G} to update reputations in homogenous resident populations. We also plot the evolution of cooperation over time in the corresponding resident populations. We show these time series for two values of pairwise continuation probability in the population, $\delta = 0.9$ and $\delta = 0.999$. In **(B)**, We plot the time average of α_{I}, α_{G} and cooperation rate in populations over long simulations for values of δ ranging between 0.5 and 0.999. The likelihood that individuals use generalized reciprocity drops with more expected rounds. The cooperation rate grows simultaneously.



Supplementary Figure S6: The distribution of strategies and transitions between them in an evolutionary simulation where individuals were only allowed to select between pure direct and pure generalized reciprocity. For this figure, evolutionary simulations were performed for a high value of pairwise continuation probability, $\delta = 0.999$. The methods behind the simulation and the related parameter values are the same as the simulation used to create Fig. 5 in main text.

Appendix: Proofs

Proof of Proposition 1. We define a function $f_{ij}(T)$ in the following way,

$$f_{ij}(T) := (1-d)\sum_{\tau=0}^{T} d^{\tau}(dx_{ij}(\tau+1) - x_{ij}(\tau)) \tag{38}$$

In the above definition we first insert the recursive expression of $x_{ij}(\tau+1)$ in terms of $x_{ij}(\tau)$ from Eq. (3) and then evaluate $\lim_{T\to\infty} f_{ij}(T)$. While doing the steps, we use the definition average reputation of j with respect to i, x_{ij}, from Eq. (8) to simplify and get,

$$\begin{aligned}\lim_{T\to\infty} f_{ij}(T) &= (d - dw - d(\bar{w}-w)(\lambda_i+\gamma_i))x_{ij} + dwr_i x_{ji} + dw\lambda_i r_i(1-2\varepsilon)\sum_{l\neq i,j} x_{jl} \\ &+ dw\gamma_i r_i \sum_{l\neq i,j} x_{li} + dwq_i + dw\gamma_i q_i(n-2) + dw\lambda_i(q_i+\varepsilon r_i)(n-2) \\ &- x_{ij}\end{aligned} \tag{39}$$

There is way to evaluate another expression for $\lim_{T\to\infty} f_{ij}(T)$. This is by observing that the sum in the definition of $f_{ij}(T)$ is a telescopic sum,

$$\begin{aligned} f_{ij}(T) :&= (1-d)\sum_{\tau=0}^{T} d^{\tau}(dx_{ij}(\tau+1) - x_{ij}(\tau)) \\ &= (1-d)(d^{T+1}x_{ij}(T+1) - x_{ij}(0))\end{aligned} \tag{40}$$

and therefore,

$$\lim_{T\to\infty} f_{ij}(T)T :\, = -(1-d)x_{ij}(0) = -(1-d)y_i \tag{41}$$

Comparing Eq.(39) and Eq.(41) we equate the two right hand sides to get,

$$\begin{aligned}&(d - dw - d(\bar{w}-w)(\lambda_i+\gamma_i) - 1)x_{ij} + dwr_i x_{ji} + dw\lambda_i r_i(1-2\varepsilon)\sum_{l\neq i,j} x_{jl} \\ &+ dw\gamma_i r_i \sum_{l\neq i,j} x_{li} + dwq_i + dw\gamma_i q_i(n-2) + dw\lambda_i(q_i+\varepsilon r_i)(n-2) = -(1-d)y_i\end{aligned} \tag{42}$$

In a homogenous population playing the strategy $\sigma = (y, p, q, \lambda, \gamma)$, $x_{ij} = x_{ji} = x_{jl} = x_{li} = x$. Then, Eq. (42) reduces to,

$$\begin{aligned}
&(d - dw - d(\bar{w} - w)(\lambda + \gamma) - 1)x + dwrx + dw\lambda r(1 - 2\varepsilon)(n - 2)x \\
&+ dw\gamma r(n - 2)x + dwq + dw\gamma q(n - 2) + dw\lambda(q + \varepsilon r)(n - 2) = -(1 - d)y \\
\implies x &= \frac{(1 - d)y + dwq + d(\bar{w} - w)(q\lambda + q\gamma + \lambda\varepsilon r)}{(1 - d) + dw(1 - r) + d(\bar{w} - w)((\lambda + \gamma)(1 - r) + 2\varepsilon\lambda r)}
\end{aligned} \tag{43}$$

Now using the expressions $w = 2/(n(n-1))$ and $\bar{w} = 2/n$ and the expression for δ in Eq.(1), one can simplify x as,

$$x = \frac{(1 - \delta)y + \delta q + \delta(n - 2)(q(\lambda + \gamma) + \lambda\varepsilon r)}{1 - \delta r + \delta(n - 2)((\lambda + \gamma)(1 - r) + 2\lambda\varepsilon r)} \tag{44}$$

Here $r = p - q$. The payoff of every player in this homogenous population is $(b - c)x$. For $0 < \delta < 1$, $n > 2$ and $\varepsilon > 0$, one can observe from the numerator of the expression in Eq. (44) that $x = 0$ (a fully defective population) if and only if either a) $y = q = 0$ and $p = 0$ or b) $y = q = 0$ and $\lambda = 0$. Here $y = q = 0$ is a necessary condition for $x = 0$, and, together with either $p = 0$ or $\lambda = 0$ it is also sufficient. For $0 < \delta < 1$, $n > 2$ and $\varepsilon = 0$, $y = q = 0$ is necessary as well as sufficient for $x = 0$. The same is true for $n = 2$.

Similarly, now if we plug $x = 1$ (a fully cooperative population) in Eq. (44), we get,

$$(1 - \delta)y = 1 - \delta p + \delta(n - 2)((\lambda + \gamma)(1 - p) + \lambda\varepsilon(p - q)) \tag{45}$$

When $n = 2$, the only solution for the above equation is $y = p = 1$. However, when $n > 2$, the left hand side in itself is always less than or equal to $1 - \delta$ while the right hand side in itself is always greater than or equal to $1 - \delta$. When $0 < \delta < 1$, the only case when they are exactly equal to $1 - \delta$ is when $y = p = 1$ and either $q = 1$ or $\lambda = 0$. □

Proof of Lemma 1. In the case where there is one mutant in a population of size labeled as player n, we denote the average probability that a resident considers another resident *good* with x_{ik}, and that a resident considers the mutant *good* as x_{in} and that the mutant considers the

resident *good* as x_{ni}. The resident plays the strategy $(y, p, q, \lambda, \gamma)$. Then, rewriting Eq. (42) from the perspective of a resident i, towards the mutant n, i.e., by replacing j with n in Eq. (42) we get,

$$\begin{aligned}&(d - dw - d(\bar{w} - w)(\lambda + \gamma) - 1)x_{in} + dwrx_{ni} + dw\lambda r(1 - 2\varepsilon)(n - 2)x_{ni} \\ &+ dw\gamma r(n - 2)x_{ik} + dwq + dw\gamma q(n - 2) + dw\lambda(q + \varepsilon r)(n - 2) = -(1 - d)y\end{aligned} \tag{46}$$

Rearranging the terms of the above equation so that the coefficient of x_{in} is 1, we get,

$$x_{in} = B_1 + B_2 x_{ni} + B_3 x_{ik} \tag{47}$$

where B_1, B_2 and B_3 are given in Eq. (13). If we rewrite again Eq. (42) but this time from the perspective of a resident i, towards another resident k, i.e., replacing j with k in Eq. (42) we get,

$$\begin{aligned}&(d - dw - d(\bar{w} - w)(\lambda + \gamma) - 1)x_{ik} + dwrx_{ik} + dw\lambda r(1 - 2\varepsilon)(n - 3)x_{ik} \\ &+ dw\gamma r(n - 3)x_{ik} + dw\lambda r(1 - 2\varepsilon)x_{in} + dw\gamma rx_{ni} + dwq + dw\gamma q(n - 2) + \\ &+ dw\lambda(q + \varepsilon r)(n - 2) = -(1 - d)y\end{aligned} \tag{48}$$

Rearranging the terms of the above equation so that the coefficient of x_{ik} is 1, we get,

$$x_{ik} = C_1 + C_2 x_{ni} + C_3 x_{in} \tag{49}$$

where C_1, C_2 and C_3 are given in Eq. (13). □

Proof of Proposition 2 . The linear equations in the previous lemma hold only if $n > 2$. In this case, combining Eq. (47) and Eq. (49) and eliminating x_{ik}, one gets for $n > 2$,

$$x_{in} = \left(\frac{B_1 + B_3C_1}{1 - B_3C_3}\right) + \left(\frac{B_2 + B_3C_2}{1 - B_3C_3}\right) x_{ni} \tag{50}$$

For the simpler case of $n = 2$, the relationship between x_{in} and x_{ni} is only due to direct reciprocity and is given by,

$$x_{in} = (1 - \delta)y + \delta q + \delta r x_{ni} \tag{51}$$

The two equations above can be rewritten using K_1 and K_2 from (15) and Eq. (16) respectively,

$$x_{in} = K_1 + K_2 x_{ni} \tag{52}$$

The payoff of the mutant π_n can be simplified using the linear relationship above in the following manner,

$$\pi_n = \frac{1}{n-1} \sum_{i \neq n} b x_{in} - c x_{ni} \tag{53}$$

$$= \frac{1}{n-1} \sum_{i \neq n} b(K_1 + K_2 x_{ni}) - c x_{ni} \tag{54}$$

$$= K_1 b + (K_2 b - c) x_{ni} \tag{55}$$

□

Proof of Lemma 2. With some algebraic steps it can be confirmed that for $n > 2$, $K_2 b - c$ can be expressed as, $(\Lambda_1 r^2 + \Lambda_2 - \Lambda_3)/(1 - B_3C_3)$. To complete the proof we only need to show that $1 - B_3C_3 > 0$. From the expressions of B_3 and C_3 in Eq. (13), it is easy to confirm that $0 \leq B_3 < 1$ and $0 \leq C_3 < 1$. Thus, $1 - B_3C_3 > 0$. The signs of $K_2 b - c$ and $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3$ are therefore always the same. □

Proof of Theorem 1. Resident strategies enforce a linear relationship on the mutant's payoff based on how cooperative the mutant is to the residents (Proposition 2). Residents can be classified into three groups: a) ones which enforce a zero slope on this relationship: $K_2 b - c = 0$ (equalizer strategies) or b) ones which enforce a positive slope: $K_2 b - c > 0$ or c) ones which enforce a negative slope: $K_2 b - c < 0$. From Lemma 2 we know that it is sufficient to check the sign of the quadratic on r: $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3$ to group resident strategies into these three classes. Let's look at each case:

When $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 = 0$: All resident strategies that equalize mutants' payoff are Nash equilibrium strategies. Since for all values of average cooperation the mutant's payoff is the same, every possible strategies is a best response to the resident. The resident strategy is therefore also a best response to itself. We classify these Nash strategies under *generic Nash*.

When $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 < 0$: Resident strategies that enforce a negative slope on the mutant's payoff relationship necessitates full defection as the only best response for a mutant against the resident. Resident strategies that fully defect in a homogenous ($x = 0$) *and* enforce a negative slope for the mutant player's payoff are thus best response to themselves. These are strategies that simultaneously satisfy, $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 < 0$ and $y = q = 0$ and additionally $\lambda = 0$ (if $\varepsilon \neq 0$). One special strategy that satisfies these properties is ALLD: $y = p = q = r = 0$. For this strategy, $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 = -\Lambda_3 < 0$. We also classify ALLD under the class of generic Nash.

When $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 > 0$: Resident strategies that enforce a positive slope on the mutant's payoff relationship necessitates full cooperation as the only best response for a mutant against the resident. Resident strategies that fully cooperate in a homogenous ($x = 1$) *and* enforce a positive slope for the mutant player's payoff are thus best response to themselves. These are strategies that simultaneously satisfy, $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 > 0$ and $y = p = 1$ and additionally $\lambda = 0$ (if $\varepsilon \neq 0$). □

Proof of Theorem 2. For $n = 2$, the linear relationship between mutant's payoff and its average cooperation to the resident (now just a single player), x_{ni} is given by,

$$\pi_n = (1 - \delta)y + \delta q + (\delta(p - q)b - c)x_{ni} \tag{56}$$

When $p - q = c/(b\delta)$, the resident is an equalizer and therefore a Nash equilibrium (using the same argument as the previous proof). When $p - q < c/b$, the strategy that always defects against itself: $y = q = 0$ is a Nash equilibrium. That is, $y = q = 0$, $p < c/(b\delta)$ is also a Nash equilibrium. When $p - q < c/(b\delta)$ a strategy that always cooperates with itself: $y = p = 1$ is a Nash equilibrium. That is, $y = p = 1$ and $q < 1 - c/(b\delta)$ is Nash equilibrium. □

Proof of Proposition 3. By definition, a generic cooperative Nash equilibrium are generic Nash strategies. So, strategies which result in $\Lambda_1 r^2 + \Lambda_2 r - \Lambda_3 \neq 0$ are ruled out. In addition, these strategies have cooperation in the homogenous population, $x \to 1$ as $\varepsilon \to 0$. In the limit of

$\varepsilon \to 0$,

$$\lim_{\varepsilon \to 0} x = \frac{(1-\delta)y + \delta q + \delta(n-2)q(\lambda+\gamma)}{(1-\delta p) + \delta q + \delta(n-2)q(\lambda+\gamma) + \delta(n-2)(\lambda+\gamma)(1-p)} \tag{57}$$

The right hand side of the above expression is 1 only if $y = p = 1$. So, generic cooperative Nash strategies need to simultaneously satisfy $y = p = 1$ and $\Lambda_1(1-q)^2 + \Lambda_2(1-q) - \Lambda_3 = 0$. □

Proof of Proposition 4. A strategy of the form $(1, 1, q^*_{\mathrm{G}}, 0, 1)$ already satisfies the first necessary condition for a generic cooperative Nash (from Proposition 3). It only needs to satisfy the second necessary condition, $\Lambda_1(1-q^*_{\mathrm{G}})^2 + \Lambda_2(1-q^*_{\mathrm{G}}) - \Lambda_3 = 0$ to be a cooperative Nash. For $\lambda = 0$ and $\gamma = 1$ (which is the case for this strategy), the quadratic on $(1-q^*_{\mathrm{G}})$ becomes only a linear in $(1-q^*_{\mathrm{G}})$ since $\Lambda_1 = b\delta^2(n-2) - b\delta^2(n-2) = 0$. The remaining coefficients Λ_2, Λ_3 are given by,

$$\begin{aligned} \Lambda_2 &= (1+\delta(n-2))(b\delta + c\delta(n-2)) \\ \Lambda_3 &= c(1+\delta(n-2))^2 \end{aligned} \tag{58}$$

Since $(1+\delta(n-2)) > 0$, the solution of $q^*_{\mathrm{G}} = 1 - \Lambda_3/\Lambda_2$ is,

$$q^*_{\mathrm{G}} = 1 - \frac{c + c\delta(n-2)}{b\delta + b\delta(n-2)} \tag{59}$$

□

Proof of Proposition 5 . A strategy of the form $(1, 1, q, 0, \gamma)$ is a generic cooperative Nash equilibrium if $\Lambda_1(1-q)^2 + \Lambda_2(1-q) - \Lambda_3 = 0$. For $\lambda = 0$, the coefficients of this quadratic are given by,

$$\begin{aligned} \Lambda_1 &= -b\delta^2(1-\gamma)(1+(n-2)\gamma) \\ \Lambda_2 &= b\delta(1+\delta(n-2)\gamma) + c\delta(1+\delta(n-2)\gamma)(1+(n-3)\gamma) \\ \Lambda_3 &= c(1+\delta(n-2)\gamma)^2 \end{aligned} \tag{60}$$

The coefficients have the following sign : $\Lambda_1 \leq 0$ $\Lambda_2 > 0$ and $\Lambda_3 > 0$. Therefore, the quadratic in $r := 1-q$: $F(r) := \Lambda_1 r^2 + \Lambda_2 r - \Lambda_3$ has the following properties:

1. $F(0) < 0$
2. $F'(0) > 0$

3. $F''(r) < 0$

Due to the above properties, if $\Lambda_1 + \Lambda_2 - \Lambda_3 = F(1) \geq 0$ then there is exactly one root of $r \in (0, 1]$, which implies exactly one root for $q \in [0, 1)$. This results in the following condition,

$$\begin{aligned}\Lambda_1 + \Lambda_2 - \Lambda_3 =&(1 + \delta(n-2)\gamma)(b\delta + c\delta - c - c\delta\gamma) - b\delta^2(1-\gamma)(\gamma(n-2)+1) \geq 0 \\ \implies& \delta(1-\gamma)((n-2)\gamma+1) \leq (1+\delta(n-2)\gamma)\left(1 + \frac{c}{b} - \frac{c}{b\delta} - \frac{c\gamma}{b}\right).\end{aligned} \tag{61}$$

We now show that, conversely, if there is only one root of $r \in (0, 1]$ then it implies that $F(1) \geq 0$ (or equivalently, Eq. (61) holds). In this case, since the quadratic $F(r)$ satisfies the above properties, a unique root $r^* \in (0, 1]$ may occur in two ways: either $F'(r^*) > 0$ or $F'(r^*) = 0$. When $F'(r^*) > 0$ and r^* is the unique root in $(0, 1]$, it implies that $F(1) \geq 0$. We now show that the degenerate case $F'(r^*) = 0$ cannot arise. We prove this by contradiction.

We consider that $\exists r^* \in (0, 1]$ such that $F(r^*) = F'(r^*) = 0$. Then, without loss of generality, one could express $F(r)$ as $F(r) = -u(r - r^*)^2$ where $u > 0$. By comparing coefficients we get

$$\begin{aligned}-u &= \Lambda_1, \\ 2ur^* &= \Lambda_2, \\ u(r^*)^2 &= \Lambda_3.\end{aligned} \tag{62}$$

This implies that $\Lambda_2^2 = -4\Lambda_1\Lambda_3$. After some simplifications, we get

$$\begin{aligned}&b\delta^2(1+\delta n\gamma)^2\left(1 + \frac{c}{b}(1-\gamma+n\gamma)\right)^2 = 4bc\delta^2(1-\gamma)(1+n\gamma)(1+\delta n\gamma)^2, \\ \implies& (2n'^2\theta^2 + 4n'\theta + 4\theta)\gamma^2 - 2n'\theta(1-\theta)\gamma + (1-\theta)^2 = 0.\end{aligned} \tag{63}$$

Here $\theta = c/b < 1$ and $n' = n - 1 \geq 2$. One can observe that the discriminant of this quadratic in γ, $\mathrm{D} := -4(1-\theta)^2(4n'^2\theta^2 + 4n'\theta + 4\theta) < 0$ thus implying that for fixed n and $c/b < 1$ there is no γ and r^* respectively in $[0, 1]$ and $(0, 1]$ such that $F(r)$ takes the form $-u(r - r^*)^2$. Therefore, a unique $r^* \in (0, 1]$ such that $F(r^*) = 0$ requires $F'(r^*) > 0$ and thus $F(1) \geq 0$.

□

Proof of Proposition 6. The proof of this proposition follows from the result of the previous proposition. If for a value of $\gamma \in [0,1]$, the strategy $(1,1,q^*,0,\gamma)$ is the only generic cooperative Nash equilibrium strategy with $\lambda = 0$, then it must be true that $F(1) \geq 0$ (the quadratic F is defined in the previous proof).

As $F(1) \geq 0$ and $F(0) < 0$, there is exactly one root of the quadratic, $r^* \in (0,1]$. If $r^* < 1$, $F(r) > 0$, for all $r \in (r^*, 1]$. Equivalently, $F(1-q) > 0$ for all $q \in [0, q^*)$, where $q^* = 1 - r^*$. The strategy $(1,1,q,0,\gamma)$, for all $q \in [0,q^*)$, satisfies $y = p = 1$ and $F(1-q) > 0$. So, by Theorem 1 (point 3), these strategies form a continuum of non-generic equilibria. Using the same Theorem, we can conclude that there is no Nash equilibrium of the form $(1,1,q,0,\gamma)$ for $q > q^*$. When $r^* = 1$, there are no non-generic Nash equilibria of this form because there is no q for which $F(1-q) > 0$. □

References

[1] L. Schmid, K. Chatterjee, C. Hilbe, and M.A. Nowak. A unified framework of direct and indirect reciprocity. *Nature Human Behaviour*, 5:1292–1302, 2021.

[2] W. H. Press and F. D. Dyson. Iterated prisoner's dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109:10409–10413, 2012.

[3] L. A. Imhof and M. A. Nowak. Stochastic evolutionary dynamics of direct reciprocity. *Proceedings of the Royal Society B*, 277:463–468, 2010.

[4] D. Fudenberg and L. A. Imhof. Imitation processes with small mutations. *Journal of Economic Theory*, 131:251–262, 2006.